The University of Manchester

Department of Earth and Environmental Sciences

EART 25102: Fife and Angus Field Course



Field days: 15 - 19 April 2026

Course Leaders:
Rufus Brunt, Steve Covey-Crump, Russell Garwood,
Margaret Hartley, Carson Richardson

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Part 1 - Logistics

1.1 Contact information, emergency numbers and accommodation

Staffing: The staff on this field course are:

Rufus Brunt	rufus.brunt@manchester.ac.uk
Steve Covey-Crump	stephen.j.covey-crump@manchester.ac.uk
Russell Garwood	russell.garwood@manchester.ac.uk
Margaret Hartley	margaret.hartley@manchester.ac.uk
Carson Richardson	carson.richardson@manchester.ac.uk

You will be provided with emergency telephone contacts for staff members before we head out into the field.

We plan to ask a student representative to set up a group WhatsApp where we can share important information such as meet locations and departure times (and share appropriate field photographs), and for emergency communications if required. We are aware that not everyone uses WhatsApp or feels comfortable providing their personal contact details to a group chat, so this communication tool is not a replacement for paying attention to verbal announcements from staff members.

In the event of an incident or emergency you should contact a member of staff who will assist you in the situation. If you require medical treatment, a member of staff will take you to an appropriate medical facility (if possible, after telephoning NHS24 on 111 for advice).

- The nearest A&E department is at Ninewells Hospital, Dundee, DD2 1UB, **01382 660 111**.
- There is a minor injuries unit (open 08:00-18:00) at St Andrews Community Hospital, Largo Road, KY16 8AR, **01334 465 656**.

For serious accidents, incidents or emergencies you should contact the emergency services on 999 and request assistance from the police, fire or ambulance services. Inform a member of staff who will assist you in the situation.

In the event of an incident you can also contact the ***University Campus Security*** on **0161 306 9966** to request advice.

Accommodation: We will be staying at the Sleeperz Hotel in central Dundee. You will be in shared twin rooms. The accommodation address is:

Sleeperz Hotel Dundee
2 South Union Street
Dundee
DD1 4BF
Tel: 01382 725 888

Breakfast will be provided by the hotel but you will need to make your own arrangements for lunch and evening meals. There is a large Tesco Extra located a 10-minute walk from the hotel.

1.2 Itinerary

	Daytime	Evening
Tue 14 Apr	Dundee: Arrive at accommodation	
Wed 15 Apr	Crawton and Arbroath	
Thu 16 Apr	St Andrews (Rock and Spindle)	
Fri 17 Apr	Pittenweem (West Braes)	Begin working on facies analysis
Sat 18 Apr	Pittenweem (Pathhead)	Begin working on poster
Sun 19 Apr	St Monan's	Complete assessed work
Mon 20 Apr	Travel to Manchester	

We may adjust the itinerary to take into account the weather conditions. We will endeavour to tell you in advance (usually the day before) if any changes are needed.

1.3 Tide times

All of the locations on this field course are tide-dependent, and some of the outcrops are completely covered at high tide. Please be prepared to be flexible so that we can work around the tides – on some days we may need to leave the hotel earlier than normal, or else work later into the afternoon/evening. We will make sure this doesn't increase the time spent working in the field: if we need to leave early, we'll also return early; and if we need to stay out late, you will receive additional time off in the morning.

	Location		HW	LW	HW	LW
Wed 15 Apr	Crawton	time	01:05	06:50	13:05	19:12
		range	3.98 m	1.25 m	4.12 m	0.72 m
	Arbroath	time	01:40	07:26	13:40	19:49
		range	4.69 m	1.52 m	4.86 m	0.94 m
Thu 16 Apr	St Andrews (East Sands)	time	02:28	08:27	14:29	20:51
		range	4.88 m	1.11 m	5.06 m	0.59 m
Fri 17 Apr	Pittenweem	time	02:54	08:56	15:03	21:27
		range	5.18 m	0.76 m	5.35 m	0.44 m
Sat 18 Apr	Pittenweem	time	03:32	09:37	15:43	22:07
		range	5.30 m	0.58 m	5.49 m	0.42 m
Sun 19 Apr	St Monans	time	04:12	10:18	16:28	22:47
		range	5.33 m	0.52 m	5.51 m	0.58 m

Tides for all locations can be accessed at https://tides.willyweather.co.uk

1.4 Equipment Checklist

General – bring every day:
Yellow field notebook
Pencils, rubber, pencil sharpener
Coloured pencils (not water soluble)
Compass-clinometer
Hand lens
Grainsize card
Mapping board with waterproof cover†

For sedimentary logging:
Logging sheets*
Measuring tape*
Acid bottle *

For compass-and-pace mapping:
30 cm ruler
A4 plain paper*

For evening work:
Mapping pens (black, green, red, blue)
A3 and A4 plain paper*
Graph paper*
Tracing paper*
Sellotape, glue*

Safety equipment – bring every day:
Sturdy walking boots
Waterproof coat and trousers
Hard hat†
Hi-vis jacket†
First aid kit*
Whistle*
Survival bag, if wanted
Means of communication e.g., phone with fully charged battery

Personal items – bring every day:
Suitable clothing for the expected weather conditions
Enough food and water for the day
Any medicines you may need
Plastic bag to carry trash generated during the day (food wrappings, etc.)

* carried by staff or provided when needed
† provided at start of trip and carried by you in the field

1.5 Health and Safety

A risk assessment for the field course has been provided on Canvas. You should familiarize yourself with the risk assessment before heading out into the field. Please ask a member of staff if you have any questions about the risk assessment.

1. General

- You have been provided with a copy of the risk assessment for the field course, including the five days spent in Fife and Angus.
- You have been given advice on safety and behaviour.
- You have completed a medical form to inform the field course leader of any health problems that may affect your safety and wellbeing (or the safety and wellbeing of others) during the field course.
- You must tell a member of staff if a previously undisclosed physical or mental health concern begins to affect you during the field course.
- For the comfort and wellbeing of non-smokers, we ask you not to smoke in the coach or in the field (except during breaks, such as lunch time).
- Alcohol must not be consumed in the field, but may be consumed in moderation in the evenings. Any substance taken during the late evening/early morning will affect your performance in the field the following day.

2. Qualitative Assessment of Risk

- Poor behaviour is often the most significant source of risk in the field.
- You must follow instructions from staff.
- Make sure you have read the course-specific hazards.
- Staff will tell you about any relevant site-specific hazards when we arrive at a new location.
- Roadside exposures and roadside access routes provide risk from traffic.
- Take care when working on the tops of steep slopes and terraces - there are often dangerous drop-offs.
- Wear a hard hat if working near cliffs or overhanging rocks. Don't work beneath another student in case they slip or dislodge something that could fall.
- Take care when working on a rising tide and be aware of your escape route.
- Wear appropriate footwear: sturdy walking boots that cover the ankles.
- Ensure you are well equipped for the expected weather conditions. Weather conditions can change rapidly.
- Carry an emergency first aid kit, or make sure you know how you can access the emergency first aid kit carried by staff.
- Prevention is key. Let a member of staff know if you have any concerns, or want to suggest a precaution that could prevent a future problem. If you are not confident ascending or descending a path or route, inform one of the staff. Don't hesitate to ask if you are unsure of anything related to safety in the field.

3. Illness

- You have completed a medical form to inform the field course leader of any health problems that may affect your safety and wellbeing or influence your work in the field.
- Should you become ill or injured during the field course, inform a member of staff immediately, even if out of normal working hours.
- Students must notify staff of any need to replace prescription medicines, such as those used to control asthma, well before the medicines are exhausted.

4. Accidents/Emergencies

- If an emergency situation arises you must inform a member of staff as soon as possible.
- Staff will be equipped with emergency first aid kits OR these will be placed in strategic positions in the field. Make sure you know how you can access emergency first aid, if needed.

5. Transport

- Do not distract the driver of a coach or minibus unless there is an urgent need to stop the vehicle.
- Equipment and luggage must be securely stowed and must not obstruct a gangway or prevent access to a door.
- Doors must be kept unlocked unless there are no passengers in the vehicle.

6. Emergencies or incidents in the field

- In the event of an emergency while we are out in the field, you should contact a member of staff using the emergency telephone contacts that we will provide at the start of the field course.

7. Emergencies or incidents out of hours

- In the event of an emergency out of hours, you should contact:
    a) University staff; and only if still unsuccessful,
    b) The duty staff at the Sleeperz Hotel.

The nearest A&E departments at Ninewells Hospital, Dundee, DD2 1UB, Tel: 01382 660 111.
There is a minor injuries unit (open 08:00-18:00) at St Andrews Community Hospital, Largo Road, KY16 8AR, Tel: 01334 465656.

Part 2 - Aims and assessment

This field course aims to develop your competence in the techniques that are essential for describing and interpreting sedimentary and igneous rocks.

You will develop skills in describing sedimentary rocks at outcrop and bed scale; interpreting sedimentary processes; and reconstructing environments and environmental change. To a lesser degree you will interpret basin-scale sedimentary sources and routing patterns. During the field course you will learn to collect datasets and use your data to address some of the fundamental questions that geologists ask themselves when mapping sedimentary rocks, such as:

- What sort of depositional environment were the sediments deposited in and what were the major processes operating in that environment?
- How does the depositional environment vary both laterally and through time?
- What sort of rocks were the sediments derived from and where was the parent rock located?
- What are the major controls on deposition within sedimentary basins?

You will develop skills in describing igneous rocks at outcrop scale, describing and interpreting igneous contacts, interpreting intrusion and extrusion processes, and assessing post-emplacement modification and alteration. You will learn to make observations and collect measurements appropriate for quantitative description of igneous rocks, and use your data to address some fundamental questions that geologists think about when mapping igneous rocks, such as:

- What is the relationship between igneous intrusions and the surrounding country rock?
- How did the igneous activity change over time, in terms of geochemical composition and intrusion/eruption style?
- What was the location or source of the igneous activity – particularly relevant for lava flows and pyroclastic deposits?

Intended learning outcomes:

At the end of the field course you should be able to:

- Collect quantitative descriptive data of sedimentary rocks, such as clast-counting and logging measured sections.
- Reconstruct sedimentary rock source area(s) and transport distances.
- Define sedimentary facies from a consideration of the types of processes responsible for the sedimentary structures, contact relations and bedforms observed.
- Group facies coherently into facies associations as a basis for inferring depositional environments.
- Analyse facies association stacking patterns to reconstruct environmental change in space and time, and provide insight into the larger scale controls on deposition within sedimentary basins such as relative and eustatic sea level changes, and the influence of tectonics on subsidence and sediment supply.
- Collect qualitative and quantitative descriptive data of igneous rocks, such as thickness of lava flows; vesicle, phenocryst or xenolith size distributions; jointing patterns.
- Reconstruct the direction to source area(s) of extrusive igneous rocks.
- Interpret mechanisms of intrusion, e.g. passive vs forceful, using observations of igneous contacts with country rock.
- Evaluate the volcanic and magmatic context of igneous activity.

Assessment

For the Fife and Angus part of the field trip there are two pieces of formative assessment, and two pieces of summative assessment.

Formative assessment

1. Your field data from St Andrews (Rock and Spindle): one fully annotated and interpreted field sketch, and short written answers in your notebook to the interpretation questions in the field guide (see Section 4.3). We won't ask you to hand this work in, but we will make sure there is an opportunity to come and discuss your work and receive formative feedback during the early evening after the St Andrews excursion. You can also ask staff for formative feedback on subsequent field days.

2. Your sedimentary log of the succession at Pittenweem (West Braes), annotated with facies analysis. We won't ask you to hand this work in, but we will make sure there is an opportunity to come and discuss your work and receive formative feedback during the early evening after the West Braes excursion. You can also ask staff for formative feedback on subsequent field days.

Summative assessment

1. An A3 poster "Evolution of sedimentary environments at Pittenweem West Braes to Pathhead". The poster should include a Geological Context, Methods, Results (summary logs, lithofacies and facies association descriptions, palaeoflow/slope indicators etc.), Interpretation (e.g. block models showing interpreted environments), and a discussion (what controlled the change in environment).

2. Your compass-and-pace map of St Monan's, and short written answers to the interpretation questions in the field guide (see Section 4.6) supported by evidence from your field notebook.

Both pieces of summative assessment will be submitted at **09:00 on Monday 20 April**, before you get on the bus to return to Manchester.

Part 3 - Geological Context

Introduction

The Devonian and Carboniferous rocks of Angus and Fife were deposited in a relatively narrow WSW-ENE-striking basin known as the Midland Valley Basin (MVB), which broadly corresponds to the central lowlands of Scotland (the "Central Belt"). The Midland Valley Basin was bounded to the north and south by two major master faults: the Highland Boundary Fault to the north, and the Southern Upland Fault to the south (Fig. 3.1, 3.2). Today, the terrain of the basin is characterised by an undulating or rolling landscape with rich arable land. The Carboniferous rocks of the Midland Valley contain the main key ingredients for steel-making (iron ore and coal – the main resource for hot furnaces, and limestone – for removing impurities) and energy generation. In the industrial revolution, beginning in the 19th Century, the Midland Valley became one of the most densely populated parts of the United Kingdom. Today it is home to close to 75% of Scotland's population (c. 4 million persons [2023]).

Geological history and stratigraphy

Detailed treatment of the geological history of the MVB can be found in Bluck (2002), Trewin and Thirlwall (2002), Read et al. (2002), and Underhill et al. (2008). This summary is derived from these and other sources.

The Midland Valley Basin is an c. 80 km-wide, >150 km-long WSW-ENE trending Late Palaeozoic fault-bounded basin, and likely extends in the subsurface westwards towards Northern Ireland and eastward into the North Sea (Fig. 3.1, 3.2). The Midland Valley Basin is bounded by Neoproterozoic rocks to the north and Lower Paleozoic rocks to the south, and contains largely an upper Palaeozoic fill. The basin as a whole has the appearance of a rift graben (Fig. 3.1, 3.2).

The Caledonian Highlands formed as a consequence of the closure of the Iapatus Ocean during the Caledonian Orogeny at the end of the early Palaeozoic. Though peak metamorphism in the orogeny occurred in what is now the Grampian region, the mountain belt extended southward as far as what is now central Wales. Beginning in the latest Silurian, a (punctuated) episode of post-orogenic extension began in the Midland Valley that initiated a protracted episode of rapid crustal subsidence and the formation of the basin. There is long-standing uncertainty concerning the plate tectonic driving mechanisms and kinematics behind this episode of extension and subsidence. The basin was originally thought to conform to a classic rift valley developed into continental crust, bounded to the NE and SE by major normal faults: the Highland Boundary Fault (HBF) to the northeast, and the Southern Upland Fault (SUF) to the southwest (e.g. George, 1960). This model was contested by Leeder (1982), who interpreted the subsidence as resulting from latest Silurian-Devonian N-S back-arc extension above the northward-subducting Gonwanan lithosphere, followed by late Carboniferous thermal subsidence. Unquestionably there is evidence of major strike-slip deformation on the major basin-bounding HBF and SUF, and along more minor intrabasinal faults, implying that the basin has a relatively complex origin and kinematic history.

At the time of early extension in the latest Silurian, Scotland lay within the arid subtropics, drifting from c. 30° south of the equator and reaching c. 20° south by the end Devonian. Initially, in the lower Devonian, the MVB was broadly segmented into two distinct elongate, parallel sub-basins, the Strathmore and Lanark basins, separated by a central chain of developing volcanoes (Fig. 3.1, 3.3,

3.4). The general picture of sedimentation in the lower Devonian was of transverse alluvial fans, delivering more locally derived sediment from the Highlands flanking the MVB to the north and south, and to a lesser extent, reworking of the volcanics that formed the sill separating the two sub-basins (Fig. 3.3). Flooring the centre of the sub-basins were broad fluvial plains, crossed by major SW-draining axial rivers probably originating in Scandinavia, and ephemeral lakes. The transverse alluvial fans were probably minor tributaries to the major axial rivers.

Abundant volcanism at around 410 Ma in Scotland is related to subduction along the northwest margin of the Iapetus (Thirwall, 1988). In the northern part of the Midland Valley, 300-1800 m of basalts, andesites and rhyolites are exposed (Fig. 3.7). Lavas and tephras intrastratify with the predominantly alluvial/fluvial clastic sediments of the ***Stonehaven Group*** and ***Strathmore Group*** (Fig. 3.5, 3.6). The Midland Valley igneous rocks are calc-alkaline in character, and include high-K shoshonitic basalts and basaltic andesites, such as the lavas of the Crawton Volcanic Formation. Some of the Midland Valley lavas show strong signatures of subducted greywacke, suggesting at least some degree of sediment melting within the subduction zone.

In the mid-Devonian, an episode of NW-SE-directed compression occurred, briefly (c. 10 Ma) inverting the MVB. Deformation was localised on the major basin-bounding faults (reverse faulting on the HBF), but also diffusely strained the lower Devonian succession through open folding (e.g. the Strathmore Syncline in the footwall of HBF; Fig. 3.4). The locus of uplift was to the south of the MVB, as evidenced by a greater lacuna in deposition to the south, and the continuation of sedimentation in the Orcadian Basin in NE Scotland and the Northern Isles (Fig. 3.3).

The Midland Valley Basin underwent renewed subsidence in the Upper Devonian, continuing into the Carboniferous. Resumption of deposition occurred via braided rivers and alluvial fans, passing upward into fluvial channel and floodplain successions characterised by the development of caliche palaeosols (the ***Stratheden Group***; Fig. 3.5, 3.8). In the Upper Devonian, major axial river systems appear to have been directed towards the NE, in direct contrast to Lower Devonian times. Aeolian dune systems formed on these floodplains too (Fig. 3.9). The entire Upper Devonian succession rests unconformably on the folded Lower Devonian succession (Fig. 3.5). In the Arbroath area there is perhaps 100 m of relief on the unconformity.

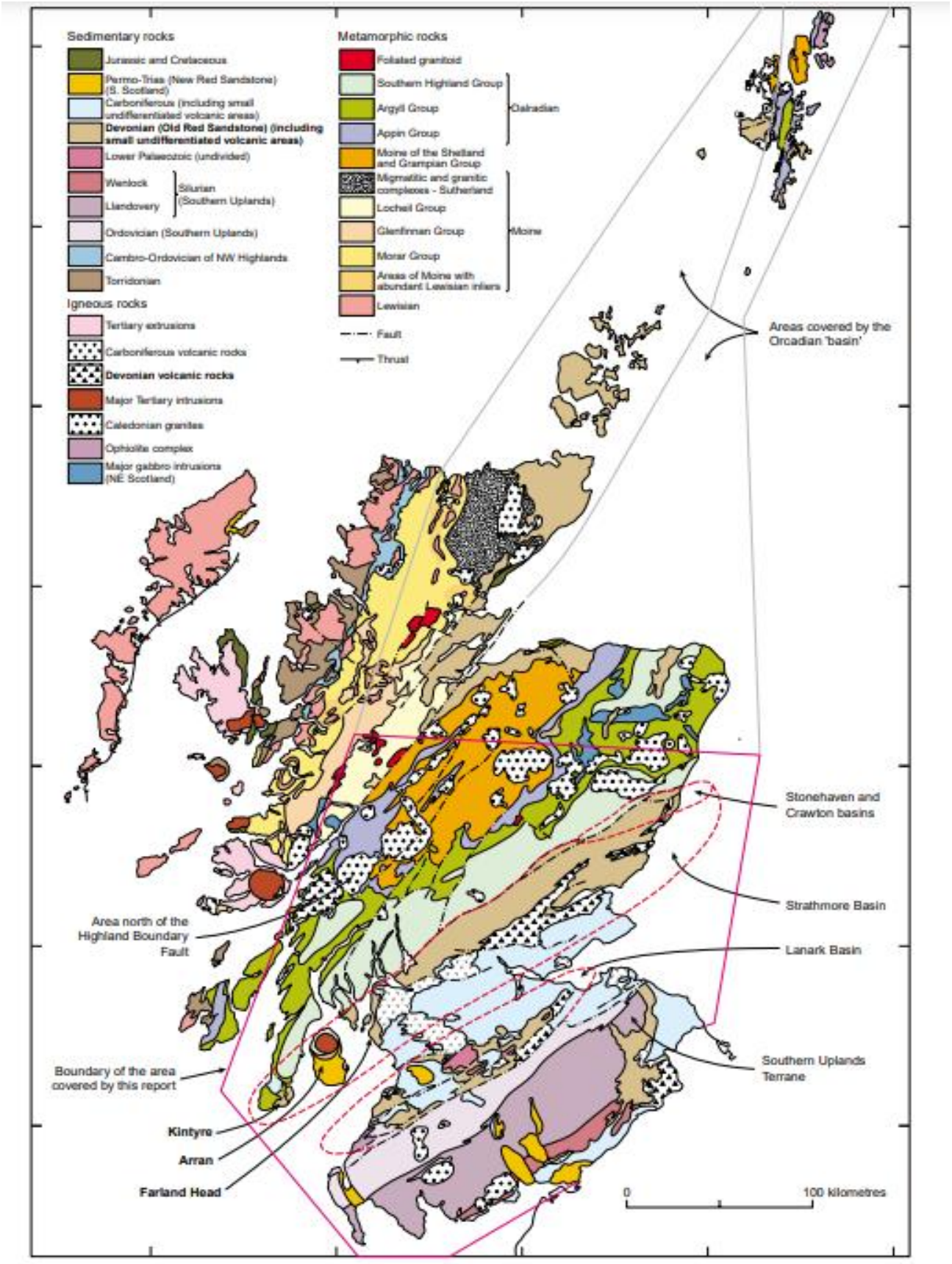


Figure 3.1: Geology of the Midland Valley within the context of Scotland as a whole.

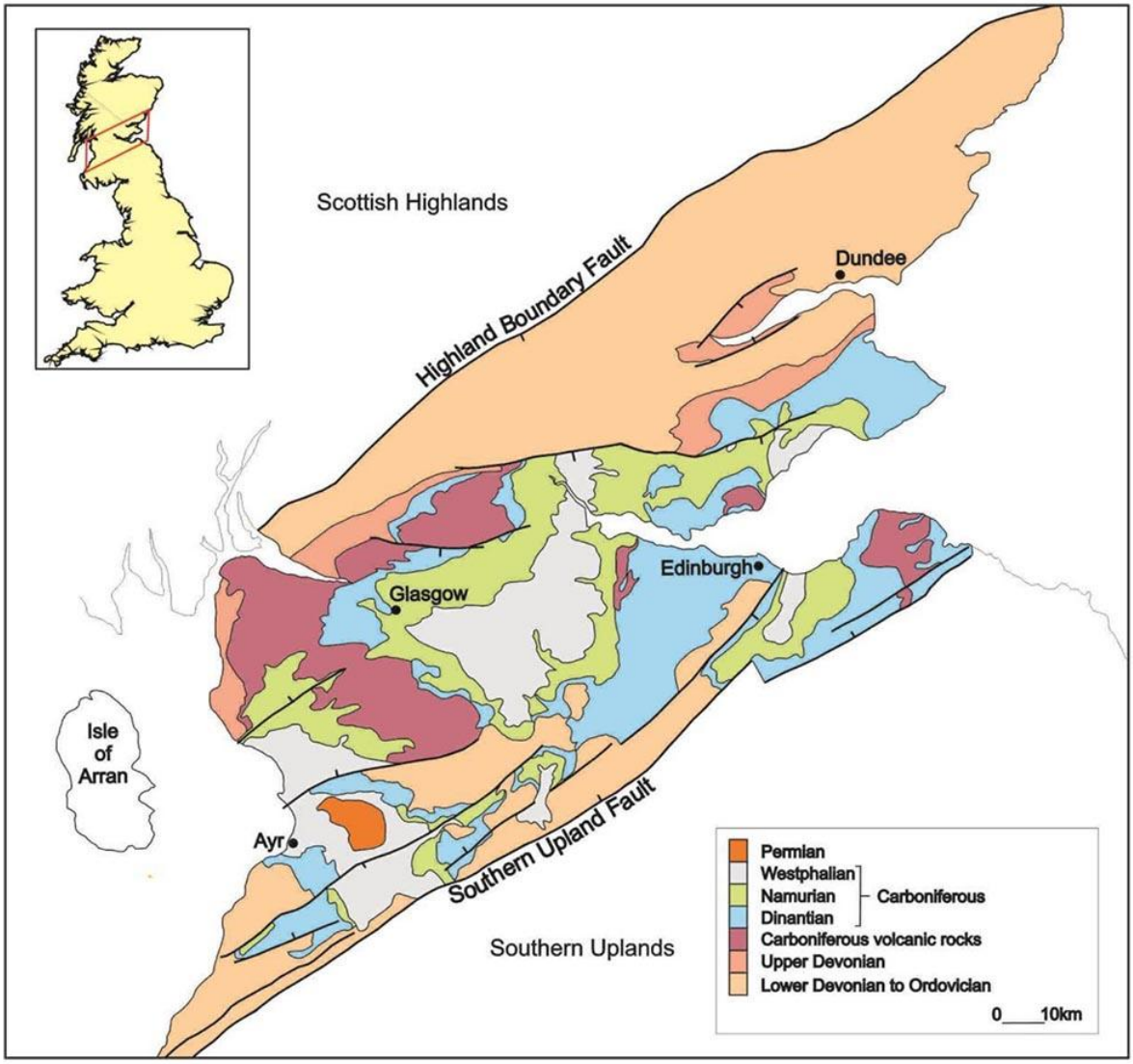


Figure 3.2: A detailed geological map of the Midland Valley Basin.

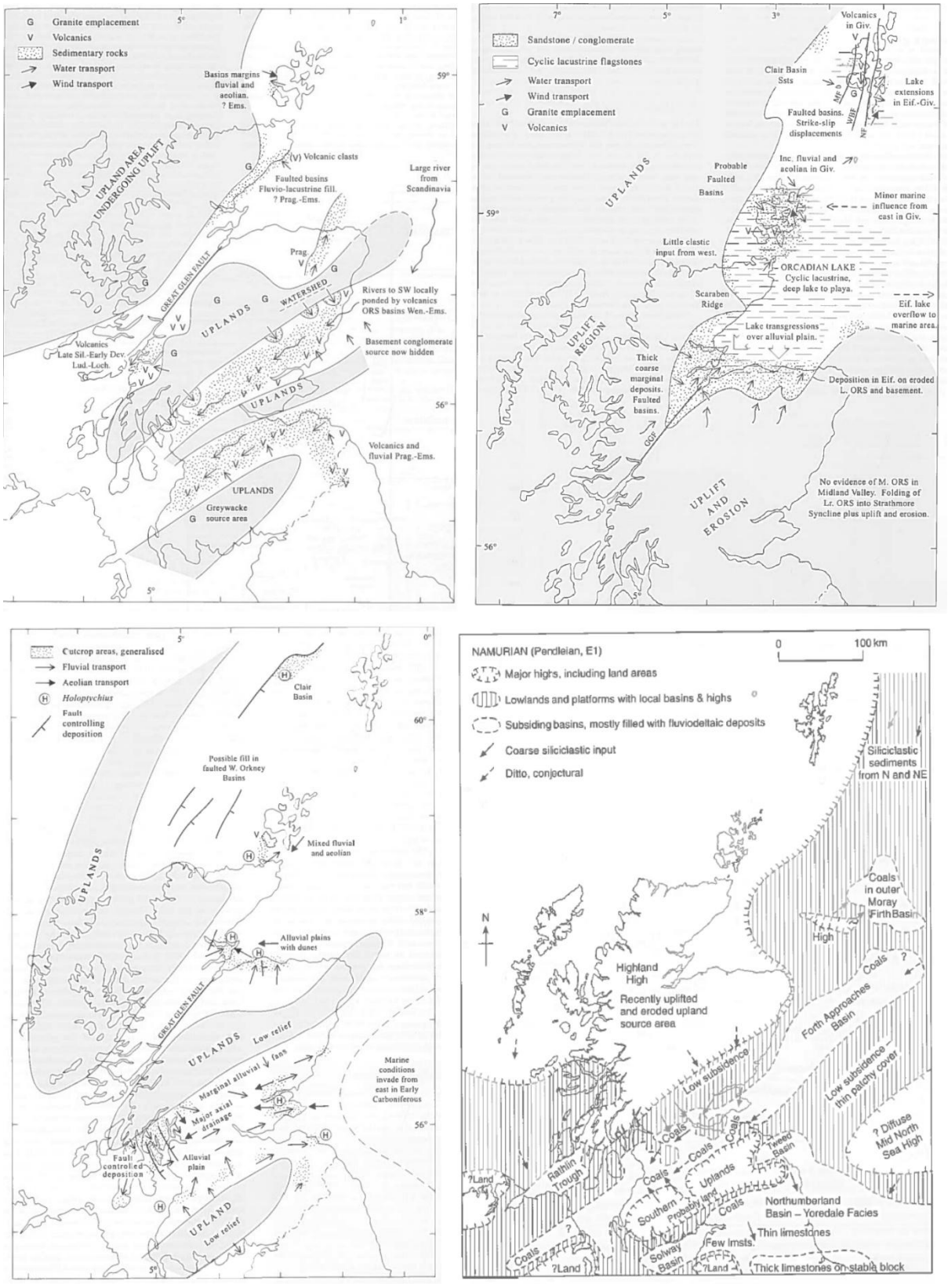


Figure 3.3: Palaeogeography of Scotland and the Midland Valley. Top left: Early Devonian. Top right: Mid Devonian. Lower left: Late Devonian. Lower right: Mid Carboniferous.

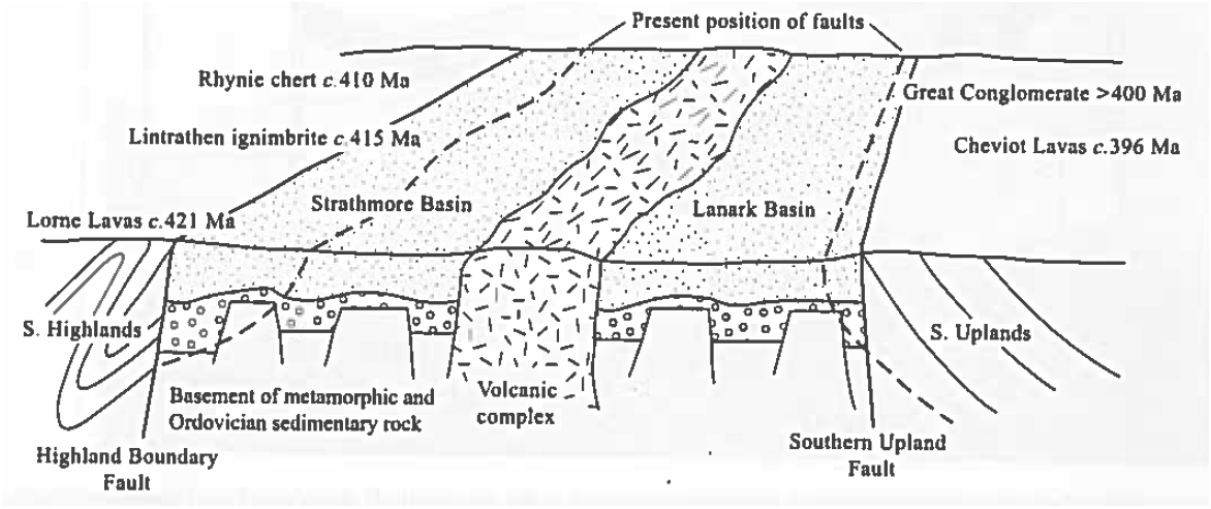


Figure 3.4: The Lower Devonian fill of the Midland Valley Basin, and illustration of the Mid Devonian contraction of the basin. The original position of the basin margin in the Lower Devonian is shown as a continuous line, and the post-deformation basin margin is shown as a dashed line.

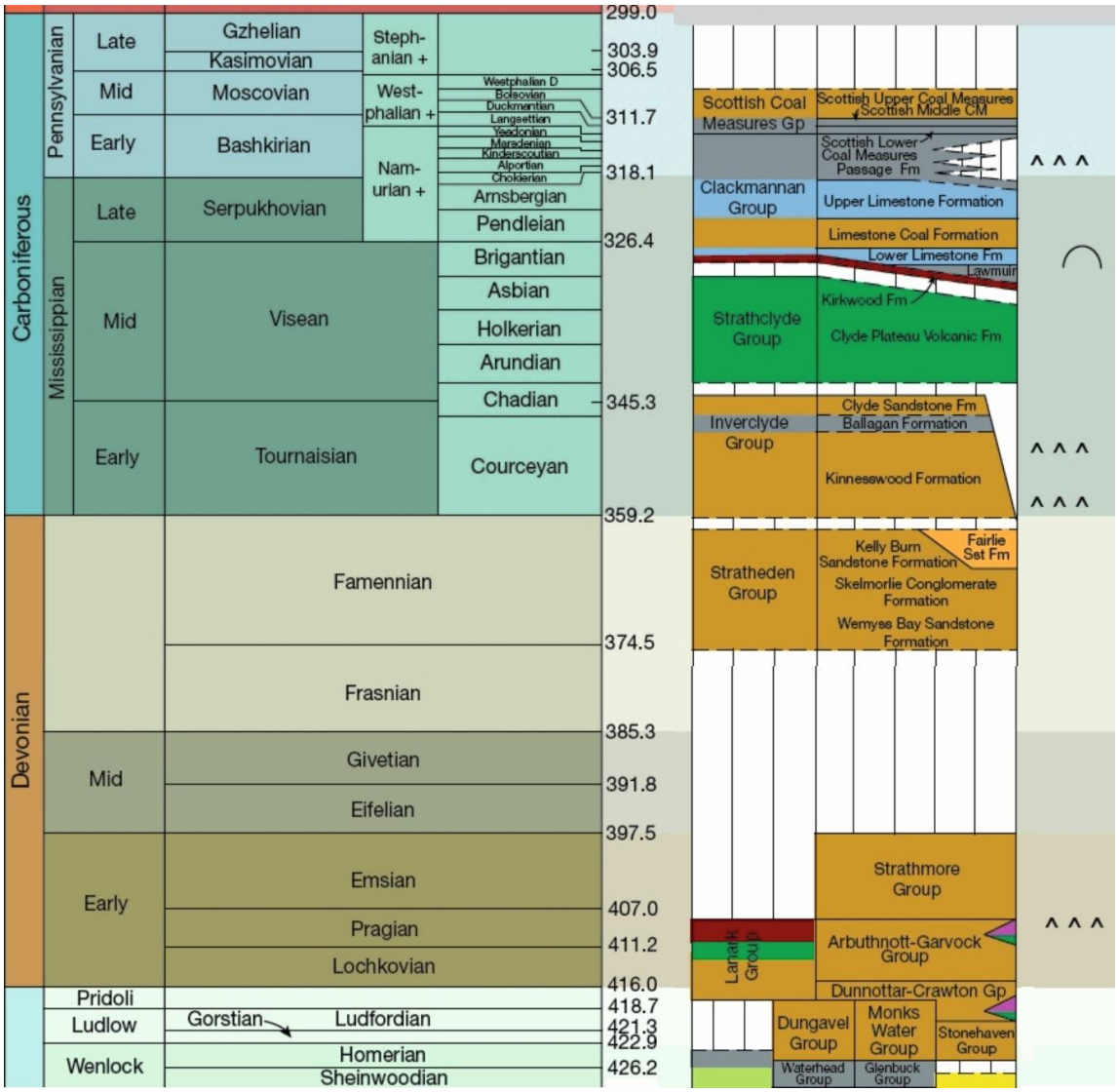


Figure 3.5: Devonian to Carboniferous lithostratigraphy of the Midland Valley Basin.

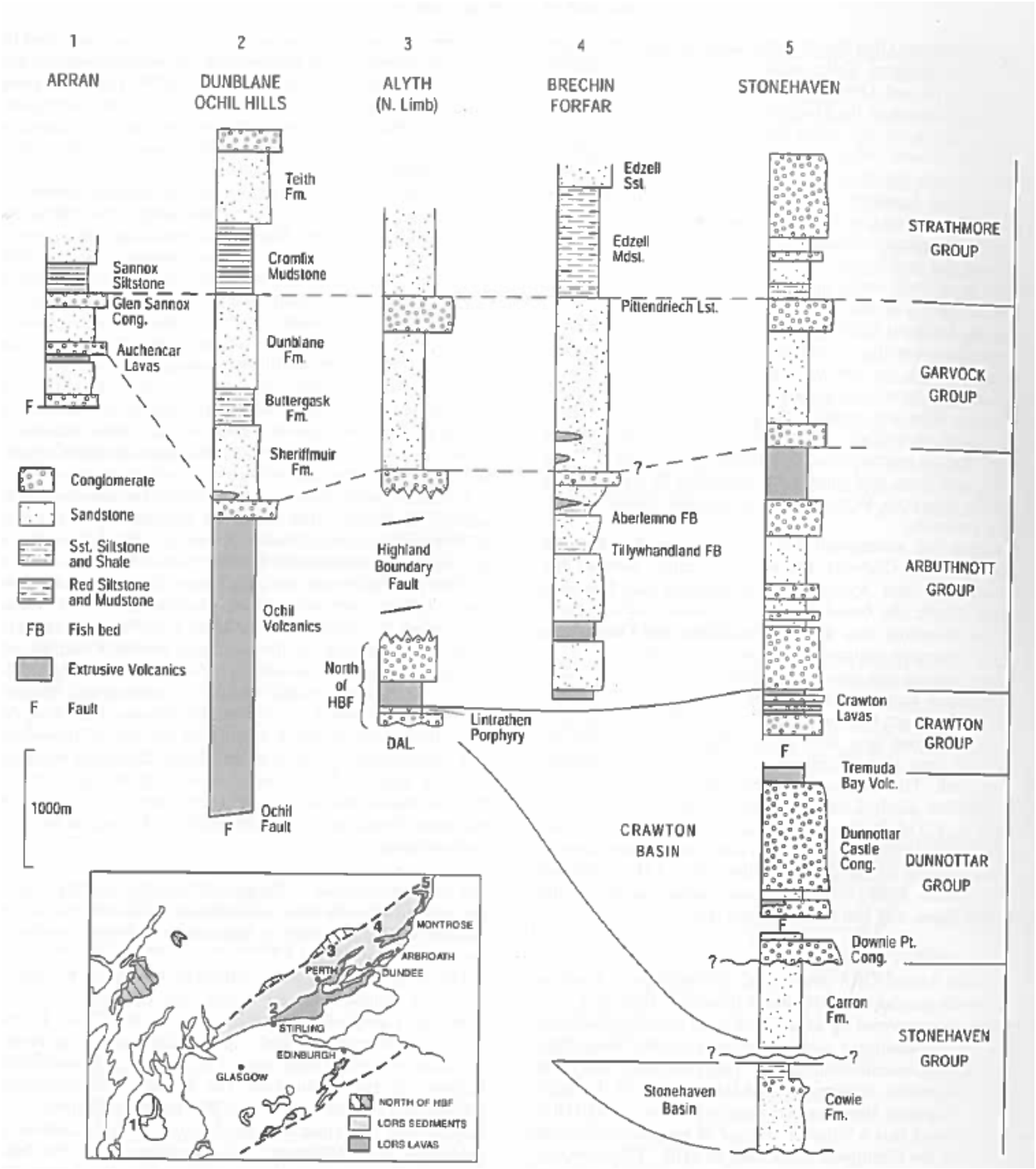


Figure 3.6: Selected sections through the Lower Devonian of the Midland Valley, highlighting facies and formation-level lithostratigraphy.

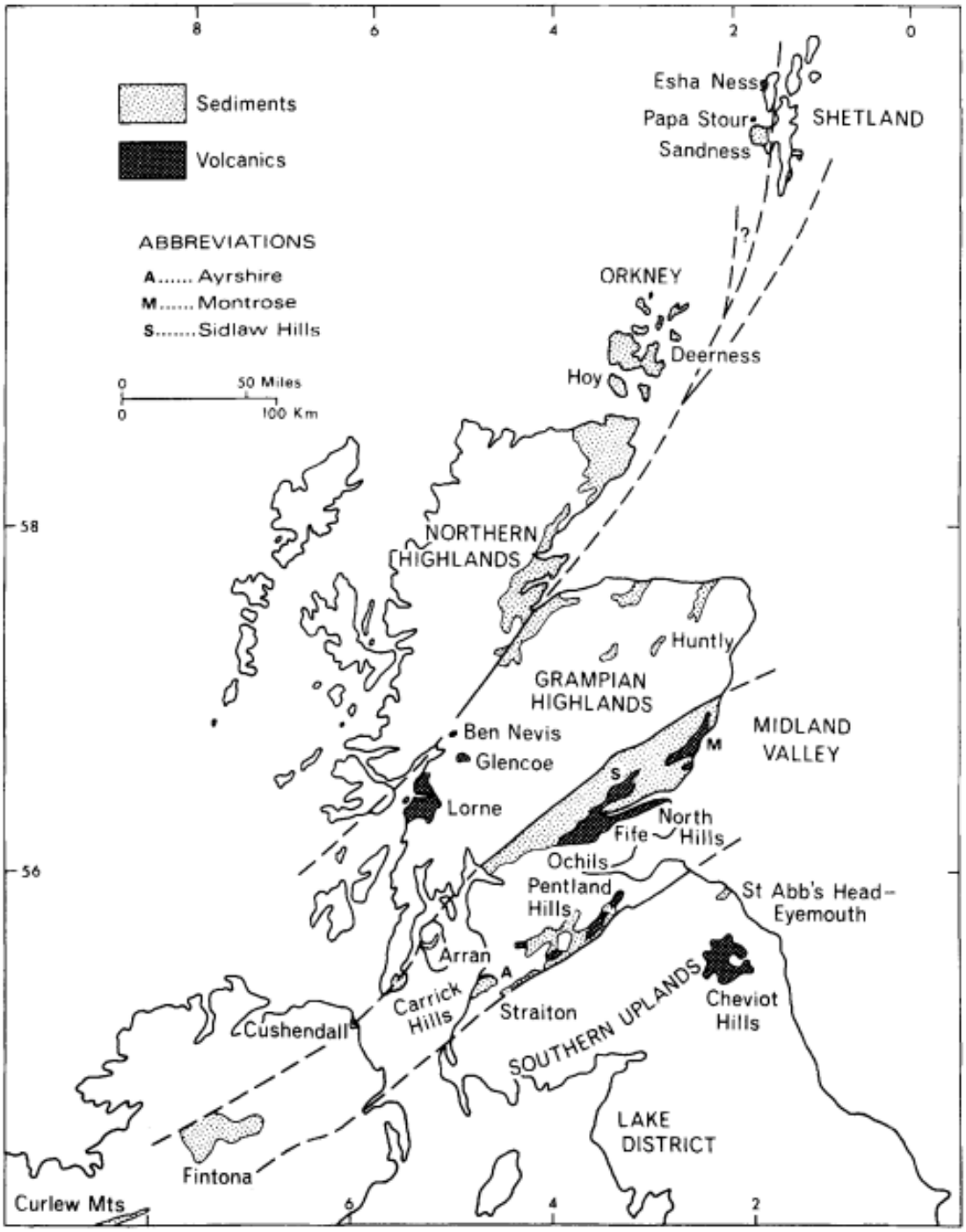


Figure 3.7: Distribution of lower and middle Old Red Sandstone (Devonian) sediments and volcanics in northern Britain.

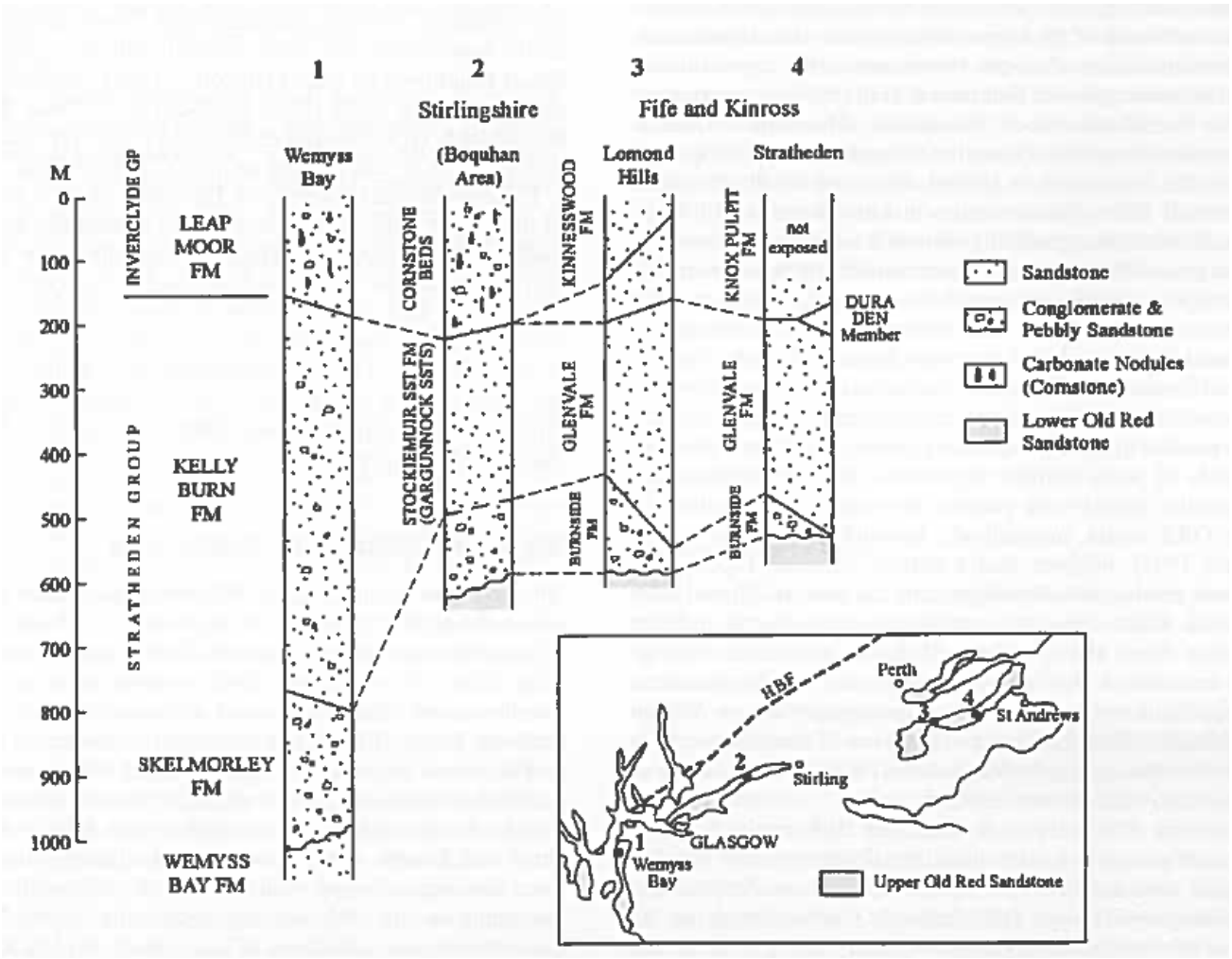


Figure 3.8: Stratigraphy of the Upper Devonian of the Midland Valley (Stratheden Group).

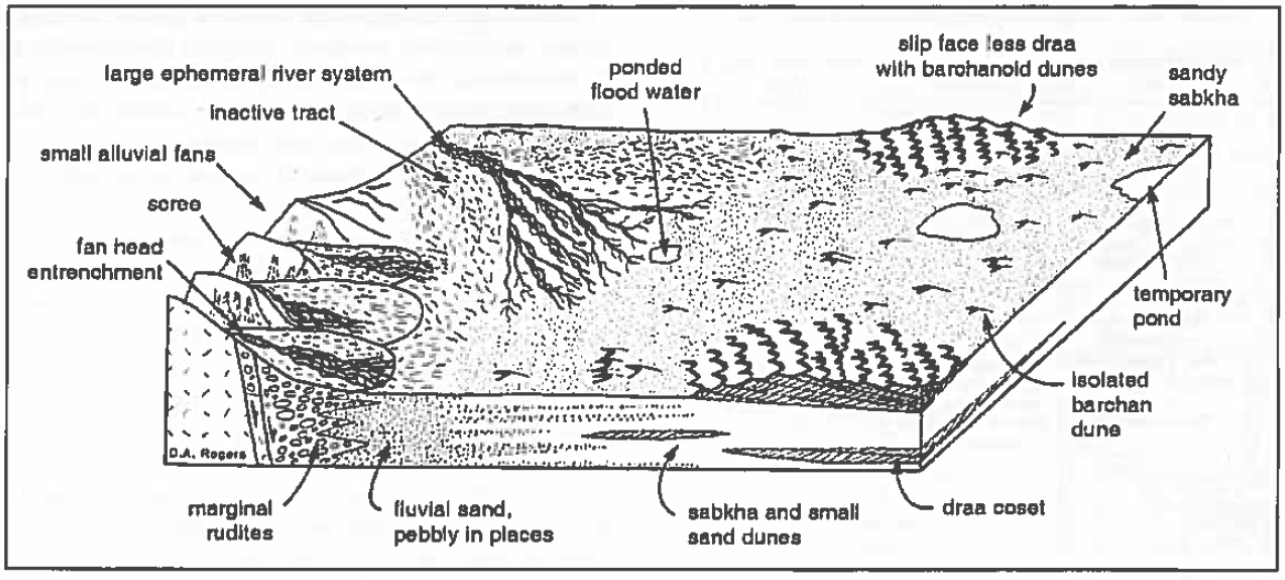


Figure 3.9: A depositional/facies model for the Upper Devonian of the Midland Valley Basin.

By Carboniferous times, Gondwana had fully docked with Laurentia to form Pangaea. A major new uplifting mountain belt, the Variscan Orogen, lay several hundreds of kilometres to the south of the MVB; the margin of the Variscan deformation lies across South Wales and Southern England. By this time the former Caledonian highlands formed much more subdued topography (but topography nonetheless). During the Carboniferous, Leeder (1982) proposes that basin subsidence in the MVB changed from fault-driven rifting to a regime of post-rift thermal sagging with lower rates of subsidence (Fig. 3.3). Nonetheless, volcanism in the basin remained active, peaking in Viséan to Namurian times. Two aspects of local and global palaeogeography are significant to deposition in the MVB. Firstly, the area drifted into the intertropical convergence zone ("the tropics"), around 10° south of the equator. This significantly altered the climatic regime in the Carboniferous to one of wet, warm climatic conditions lacking in, or with subdued, seasonality. Secondly, the Gondwanan sector of Pangaea had drifted over the palaeo-south pole, where major continental glaciation developed on a similar scale to Antarctica today. Waxing and waning of the continental icecap, possibly at Milankovic timescales, resulted in major sea level fluctuations exceeding 100 m.

The Carboniferous succession of the MVB (Fig. 3.5, 3.10) is characterised by cyclothemic deposition. Cyclothems are cycles a few metres to tens of metres thick, characterised by an upward transition through one or more of the following: (1) marine or lake mudstones, limestones and ironstones; (2) deltaic heterolith and sandstones; (3) fluvial heterolith and sandstones; and (4) coal. The idealised cycle described here would represent, at its base, a rapid deepening event (flooding), followed by gradual infilling of the bathymetry by sediment supply ultimately delivered by rivers. The base of the next cyclothem represents renewed flooding of the land surface. Cycles are rarely complete, and formations within the ***Inverclyde to Coal Measures Groups*** (Fig. 3.10) are defined according the proportion of marine, deltaic and fluvial facies present. Researchers disagree on the precise driving mechanisms for the formation of this highly cyclic sedimentation style, some invoking glacio-eustatic sea level fluctuations; others autocyclic delta switching in an overall subsiding regime; others pulses of subsidence; and yet others a combination of one, two or three of these drivers.

The rifting associated with the opening of the Midland Valley generated significant magmatic activity (Fig. 3.11). Alkaline (silica-undersaturated) and tholeiitic (silica-saturated) magmas were generated by variable degrees of partial melting of the asthenspheric mantle (Upton et al., 2004). In the Late Carboniferous, significant volumes of tholeiitic lava were emplaced, especially in the southern Midland Valley (e.g. the ***Bathgate Group***; Fig. 3.12). Alkali basaltic magmatism persisted into the Early Permian. The alkali basalts and dolerites are variously composed of plagioclase, clinopyroxene, olivine, amphibole, and analcite (a feldspathoid mineral). As the sedimentary basin fill increased, sill intrusion tended to dominate over lava extrusion. The sills, diatremes and volcanic plugs intruding the Carboniferous sedimentary succession form prominent topographic relief in the modern landscape of the Midland Valley, most famously expressed by Arthur's Seat in Edinburgh. Some of the igneous rocks are only a little younger than the sediments they intrude; in some locations there is evidence that they perhaps intruded into almost unconsolidated and still water-saturated sediment. The locations of the Fife volcanic vents were to some extent controlled by the existing basement structure: for example, along the Ardross Fault. The remains of the plumbing systems of these volcanoes remain in the rock record as discrete volcanic plugs and necks, such as at the Rock and Spindle. Many of the Fife volcanic plugs contain xenoliths of mantle peridotite and pyroxenite, as well as lower-crustal granulites and meta-igneous rocks.

The final chapter in the story of the Midland Valley Basin is of its Permian inversion. By the end Carboniferous, the basin had drifted north of the equator, and once again into the arid subtropics. This is expressed by the absence of coal and renewed development of red-beds in the ***Upper Coal***

Measures Formation (Fig. 3.10, 3.12). The Carboniferous succession grades into a short succession of Permian fluvial red beds, followed by an unconformity. True Variscan deformation and uplift was limited to South Wales and southern England (the latter now buried). However, the Upper Devonian to Carboniferous succession of the MVB was folded and thrusted, and a clear mechanism for the deformation other than the far-field effects of Variscan stress is lacking. This is nothwithstanding the absence of folding in the intervening area of Carboniferous in northern England.

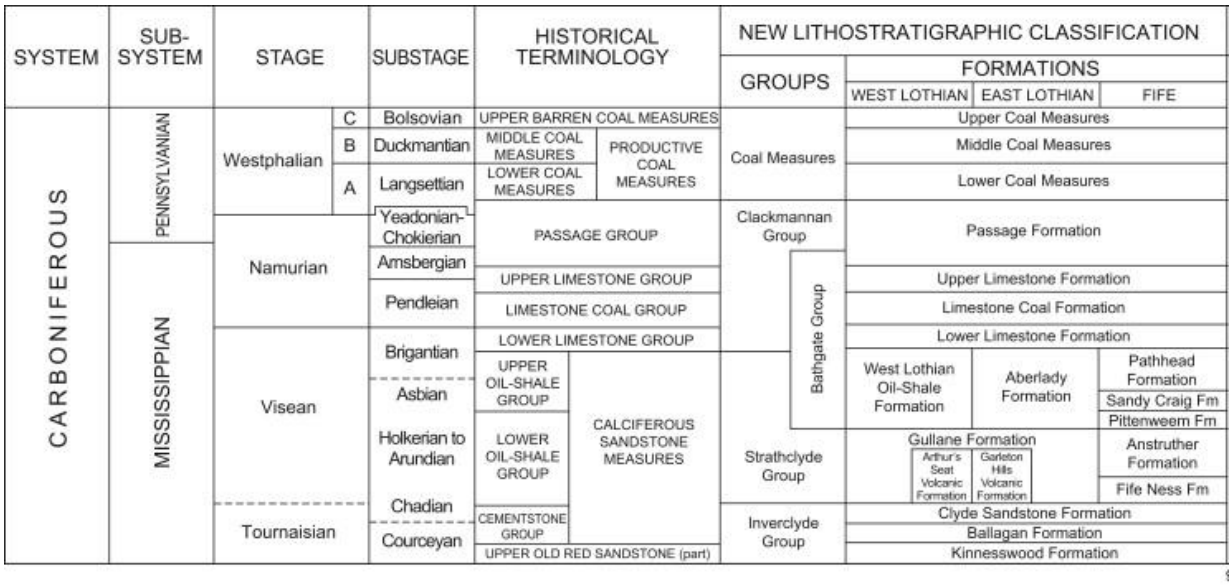


Figure 3.10: Carboniferous stratigraphy of the Midland Valley Basin.

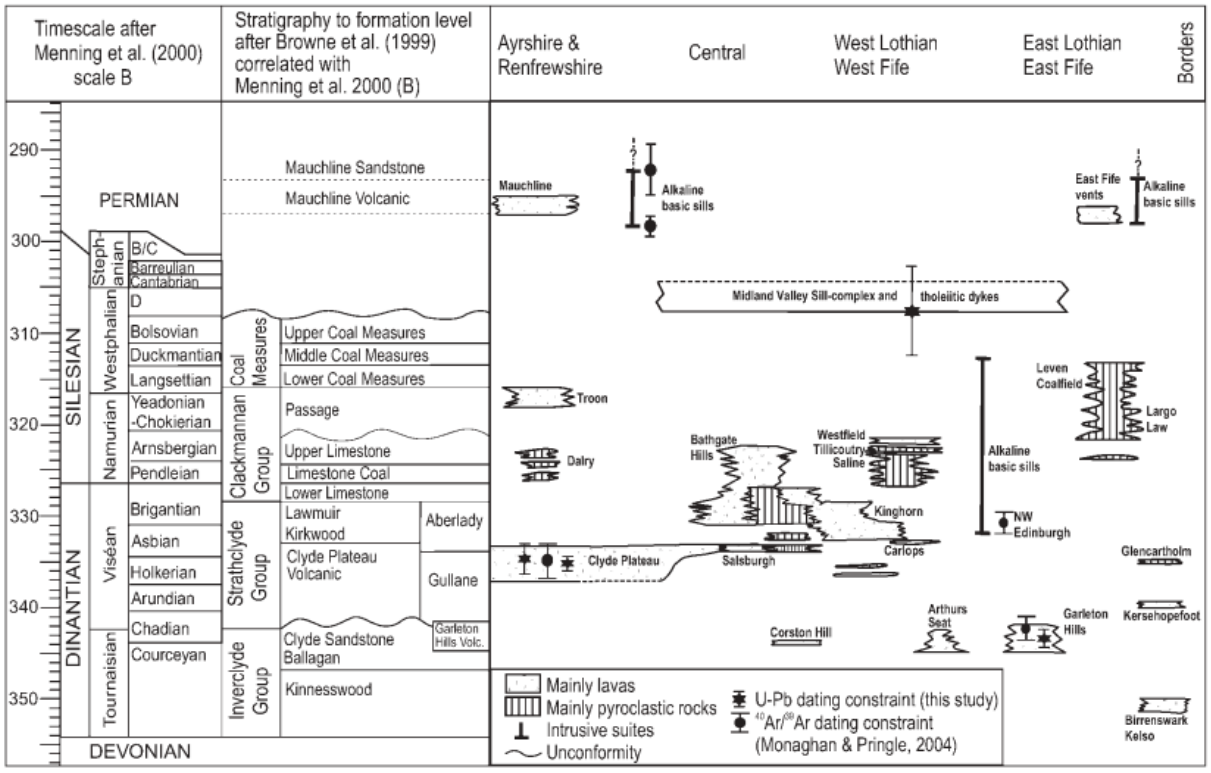


Figure 3.11: Time-space chart of Carboniferous to Early Permian volcanism in the Midland Valley.

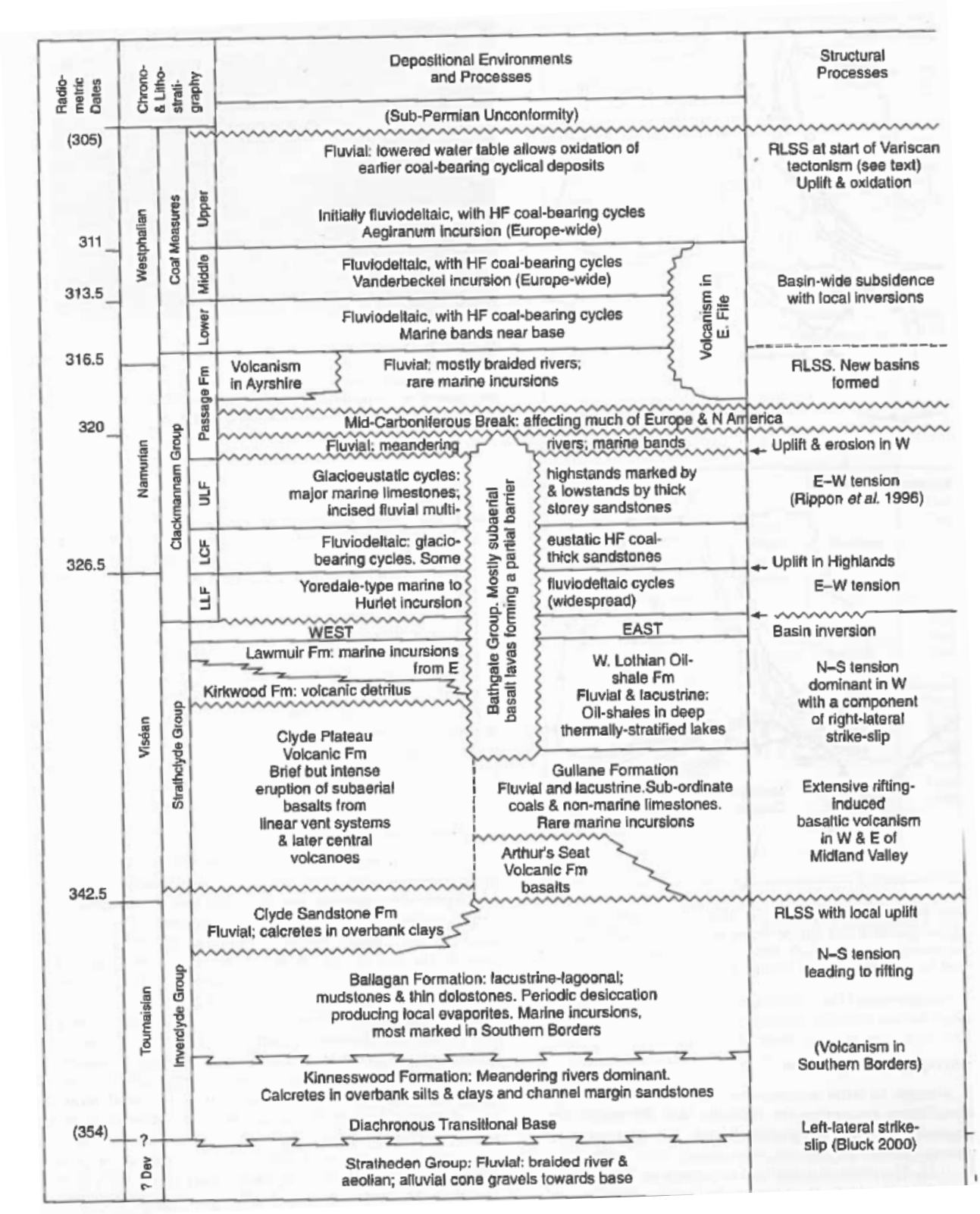


Figure 3.12: Carboniferous depositional environments, volcanic events and tectonic history in the Midland Valley.

Part 4 - Stop Information

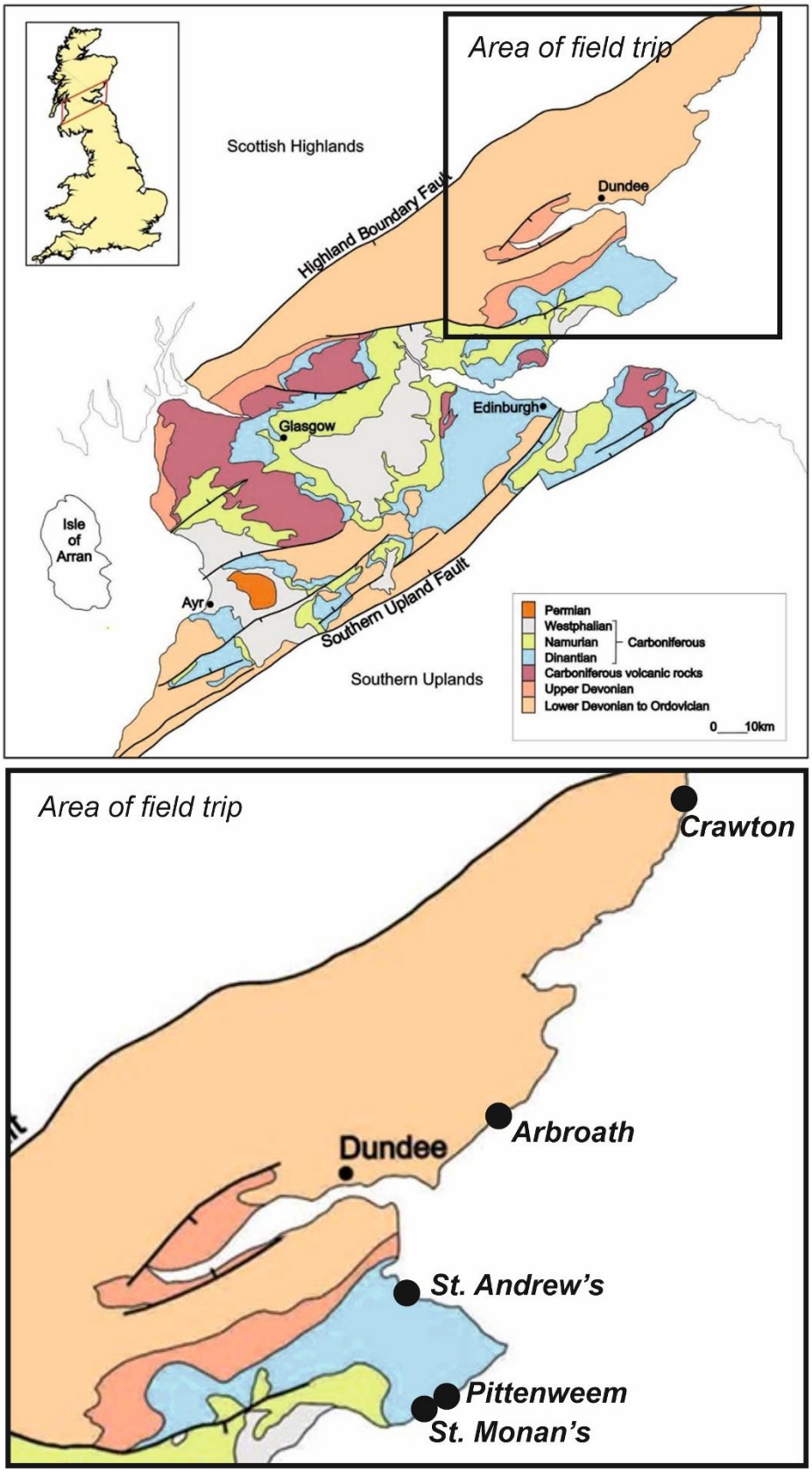


Figure 4.1: Simplified geological map of the Midland Valley Basin, with (inset) location of field stops.

4.1 Crawton

Access: Turn off the A92 along a single track road with passing places to Crawton. There is a small car park near the end of the road with turning space for a minibus if not too congested. Alternatively, disembark from vehicle at the A92 junction and walk along the single track road to Crawton (0.9 miles, 17 minutes). The exposures of interest extend across the Crawton shoreline.

There are no public toilet facilities at this location.

Geological context: The exposed rocks belong to the Crawton Volcanic Formation, which lies at the top of the Dunnottar-Crawton Group. The rocks are of Lochkovian (Early Devonian) age, ~393-444 Ma.

Aim: Develop skills for describing and measuring igneous rocks (lava flows) at outcrop scale. Develop skills for describing and measuring conglomeratic sedimentary successions. Interpret the source area(s) of the igneous and sedimentary rocks at this locality.

Tasks:

1. There are (at least) two rock types exposed at this locality. Determine the younging direction. Produce a brief rock description for each rock type you identify. Describe the nature of the contact(s) between these rock types.

2. Produce an annotated field diagram of the internal features of a lava flow in cross-section. Pay attention to vertical variations through the lava, and to horizontal variations on the scale of metres to tens of metres.

Interpretation:

- Comment on any changes in the size, shape and abundance of the vesicles with height through the lava. What causes the variations you observe?
- What direction was the lava likely flowing? What is the evidence for this?

3. Describe and sketch the upper surface of the lava flow, paying attention to the geometry of the joints.

Interpretation:

- The joints on the rock surface have weathered in an unusual way. Why do you think this might be?
- Was the lava been eroded prior to or during deposition of the conglomerate? What is the evidence for this?

4. Describe and sketch the base of the lowermost lava flow and its contact with the underlying lithology.

Interpretation:

- What direction the lava was flowing in? What other evidence have you seen that might support your interpretation?
- What is the nature of the contact? What has happened to the underlying lithology?

5. Conglomeratic units interstratify with the lava flows. A useful way of collecting data on conglomeratic successions for interpretation is to look at the bed-scale structure. Are these normally or inversely graded, or lacking grading altogether? What does this tell us about flow properties? Palaeoflow information can be determined from clast imbrication, which gives information about source direction. Clast composition, grainsize, angularity and sphericity data can be determined quantitatively by undertaking a clast count.

Summary/review:

1. How many lava flows are exposed at Crawton Bay? Are the lavas identical – consider their thickness, internal structure, grain size, crystal abundance, flow direction, vesicle size and abundance?

2. Undertake a statistical analysis (description) of the conglomerates (clast types, size, angularity and sphericity). Where did the clasts come from, and from how far away did they travel? What is your interpretation of the depositional processes and environment for these sedimentary rocks? What is their relationship to the lavas?

3. Propose and sketch a facies model for the lavas and conglomerates.

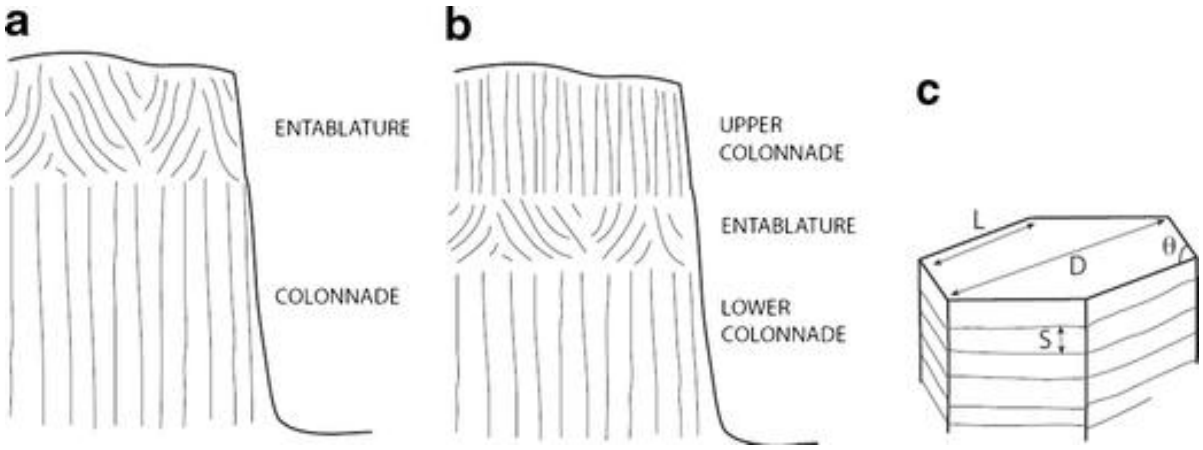


Figure 4.2: Schematic diagram of columnar jointing architecture. (a) and (b) Possible relationships between entablature and colonnade. (c) Measurable dimensions of an individual column include the diameter d, side length L, striae spacing or width S, and the internal angles ϑ. From Philips et al. (2013).

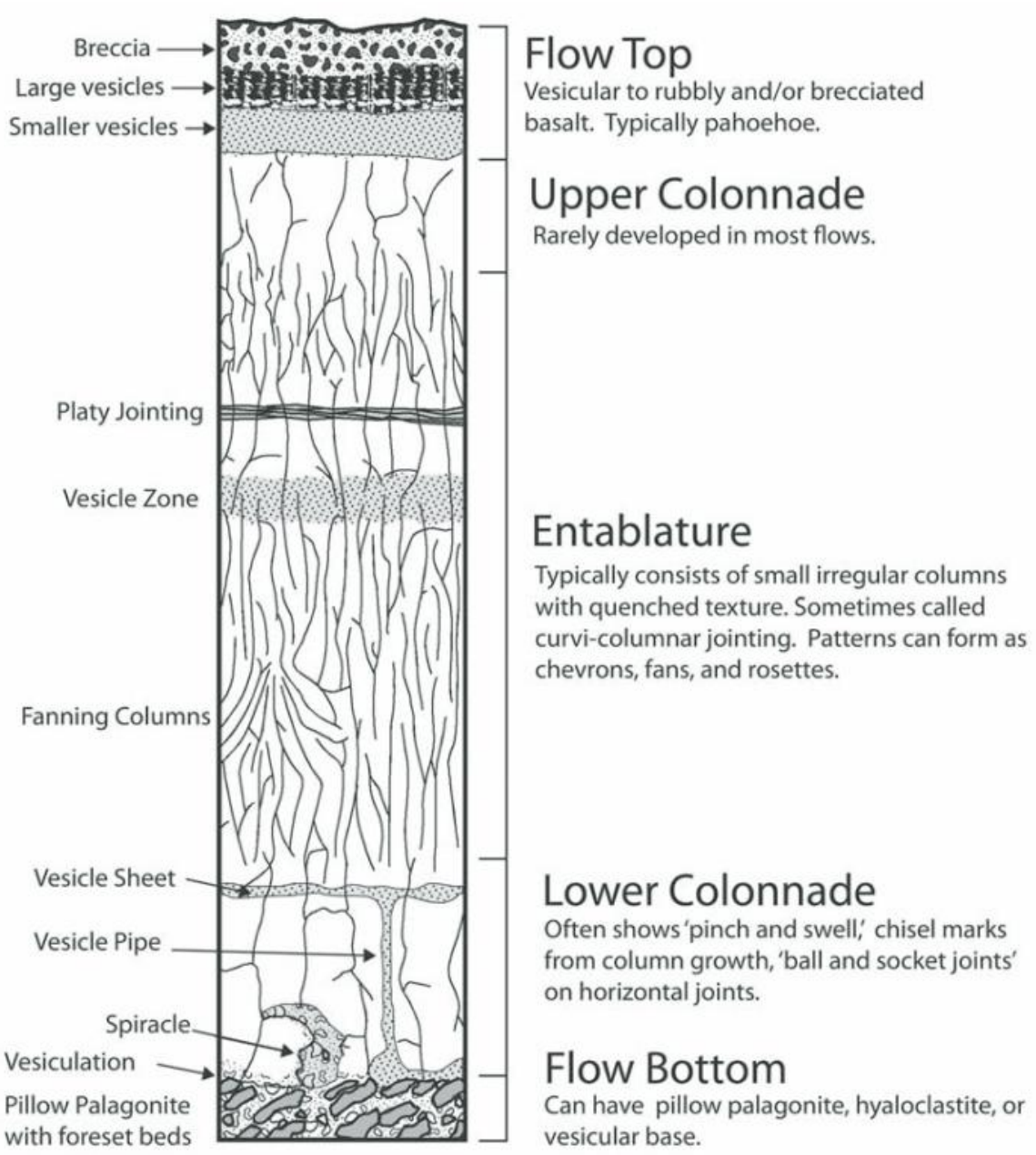


Figure 4.3: Diagram showing the typical internal features of a flood basalt lava flow. From Reidel et al. (2013).

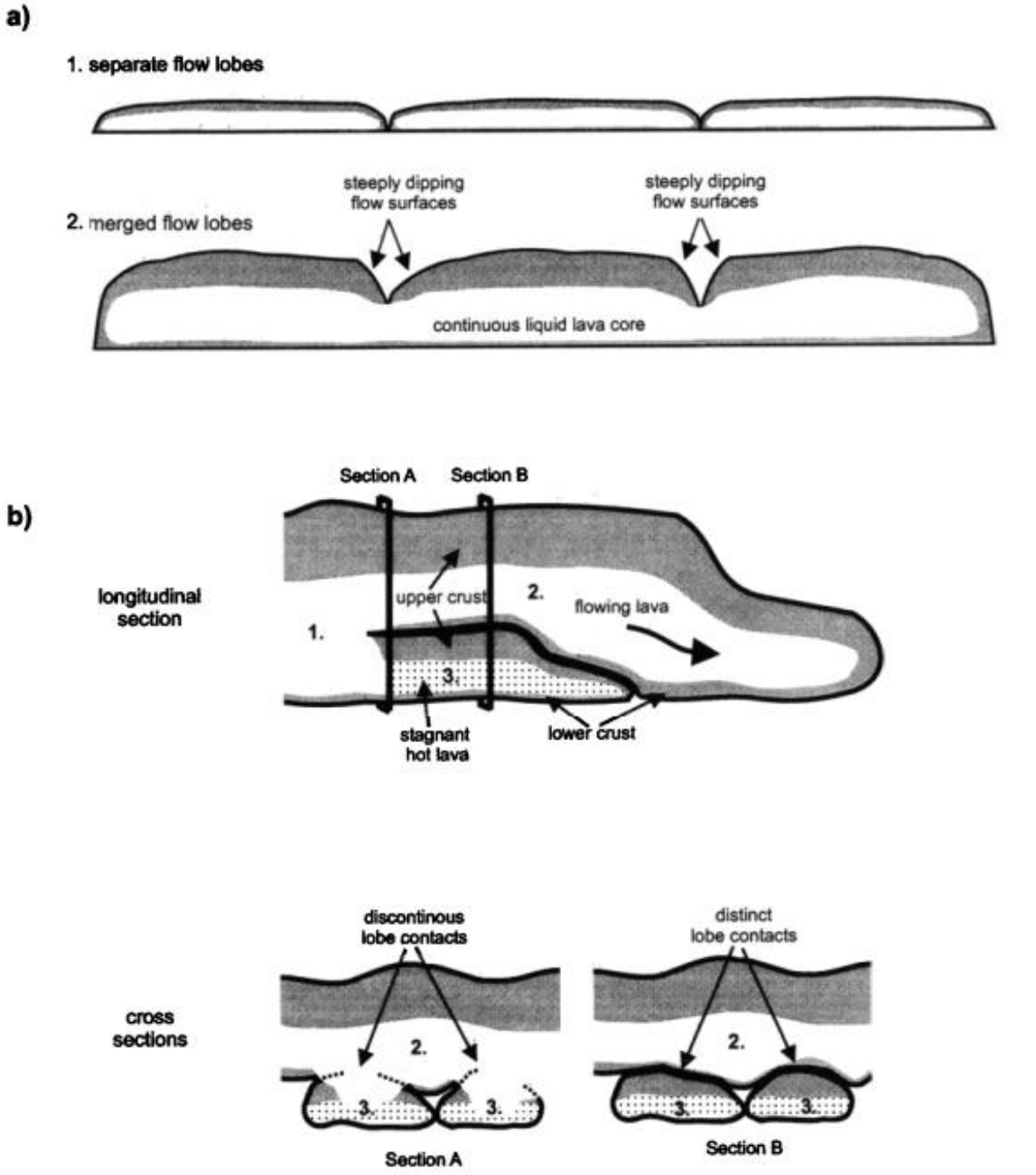


Figure 4.4: (a) Sketch illustrating the evolution of three abutting flow lobes. The flow direction is out of the page. At Stage 1, each flow is bounded by distinct surfaces on all sides. Each lobe has an upper and lower crust (shaded grey) and a liquid lava core (white). The lobes inflate and spread. By Stage 2, the lobes have merged into a single lobe due to coalescence of the liquid lava cores. The resulting sheet lobe has a continuous lava core, and steeply dipping flow surfaces that terminate near the top of the core. (b) Sketch showing an inflating sheet lobe overtaking an earlier emplaced, but now stagnant, frontal breakout. Depending on the location and orientation of the vertical cross-section through the outcrop, lobe boundaries can occur either as discontinuous contacts (cross-section A) or distinct flow surfaces (cross-section B). From Thordarson and Self (1998).

4.2 Arbroath

Access: Park at Arbroath Cliffs car park at the eastern end of King's Drive. Walk to the eastern end of King's Drive and descend the steps onto the rocky shoreline. Walk ca. 150 m eastwards along the rocky shore. The exposure of interest is the south-facing cliffs.

NB. The rocky shore is not accessible at high tide, but there is a suitable accessible (though small) exposure at the eastern end of King's Drive if arriving at high tide.

There is a public toilet at Arbroath Cliffs car park.

Geological context: The exposed rocks belong to the Scone Sandstone Formation (Arbuthnott-Gavrock Group) and the Burnside Sandstone Formation (Stratheden Group), of Devonian age.

Aim: Interpret the nature of a geological contact. Interpret changes in depositional environment across a geological contact.

Tasks:

1. Produce an annotated field sketch of the cliff and beach exposure.

2. Produce a brief rock description for each rock formation exposed at this locality. You should record information on bed-scale (within-bed) variation in grainsize; bed geometry; occurrence of sedimentary structures within the bed, on the bed base (sole structures) and bed tops; palaeoflow indicators; and anything else of interest or relevance.

Summary/review:

1. What is the nature of the geological contact at this locality? Comment on the environments of deposition of the two rock formations, noting any similarities or differences between these environments.

4.3 St Andrews (Rock and Spindle)

Access: Park at the Castle golf course, off the A917. Take the footpath from the car park to join the coastal path. Turn left onto the coastal path to access the rocky/sandy beach.

Toilets are available at the Castle Course clubhouse.

Geological context: The exposed sedimentary rocks belong to the Pittenweem Formation (Strathclyde Group) of Asbian (Viséan, Lower Carboniferous) age. The igneous rocks are of Late Carboniferous to Early Permian age (272-311 Ma).

Aim: Develop skills for describing and measuring igneous rocks on outcrop scale. Use observations to interpret mechanisms of igneous intrusion, extrusion and deposition of volcaniclastic material.

Tasks:

1. Produce an annotated field sketch of the Rock and Spindle.

Interpretation:

- What are the radial features on the "spindle"? What do they tell you about the thermal history of this rock?

2. Examine the rocks of the Rock and Spindle in closer detail. You should notice at least two different distinctive types of igneous rock. Produce a detailed rock description of the chaotic, clast-rich lithology. What clast types can you identify?

3. Examine the dyke at the base of the Rock and Spindle. Sketch the key features, and describe and classify the rock type. What phenocrysts can you identify? How does the crystal size change across the width of the dyke?

Interpretation:

- What is the age relationship between the chaotic clastic igneous rocks and the crystalline dykes? How many episodes of igneous activity can you identify?

4. Explore the area around the Rock and Spindle, particularly towards the grassy area at the edge of the beach. You should find two different lithologies here. What is the age relationship between the dark black/green rock and the paler rock? What is the composition of the matrix of the dark black/green rock? What about the grey-green clasts?

Interpretation:

- Did you interpret the dark black/green rock an intrusive or extrusive igneous rock? Is the contact an intrusive or extrusive contact? What does the lack of bedding tell you about this rock? At what depth within an igneous or volcanic system might this rock have formed?

4. Examine the bedded pyroclastic deposits that surround the smaller sea stack 15 m north of the Rock and Spindle. What is the younging direction? What is the angle of bedding? Follow one bed laterally over several metres and comment on the geometry of the layering.

In pairs or threes, compile a bed-scale (that is, at a scale where details within individual beds can be determined) log of the pyroclastic deposits. You will be guided to choose an appropriate interval to log. When choosing your logging scale, estimate the likely length of the log, and the level of detail you wish to capture. Information to record includes (but is not limited to): bed-scale (within-bed) variation in grainsize; bed geometry; occurrence of sedimentary structures within the bed, on the bed base (sole structures) and bed tops; and palaeoflow indicators.

Interpretation:

- How many discrete volcanic events are represented by the volcaniclastic stratigraphy?
- Are these pyroclastic fall deposits or pyroclastic density current (PDC) deposits? How would you interpret the flow properties of the deposits?

Summary/review:

1. Construct a schematic diagram (or series of diagrams) to summarize the igneous and volcanic processes that formed the Rock and Spindle and surrounding outcrops.

2. Comment on how you might go about working out the chemical composition of the chaotic clastic igneous rocks of the Spindle. Why might this be challenging to accomplish?

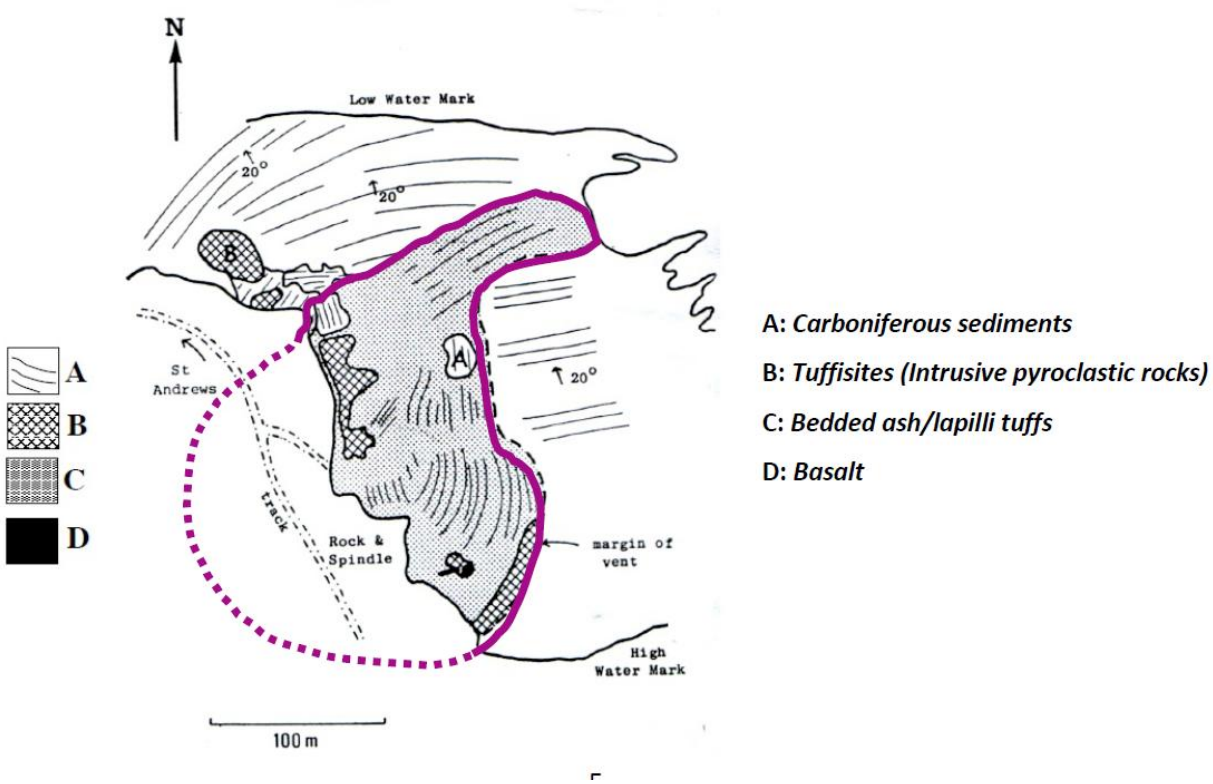


Figure 4.5: A simplified geological map of the Rock and Spindle.

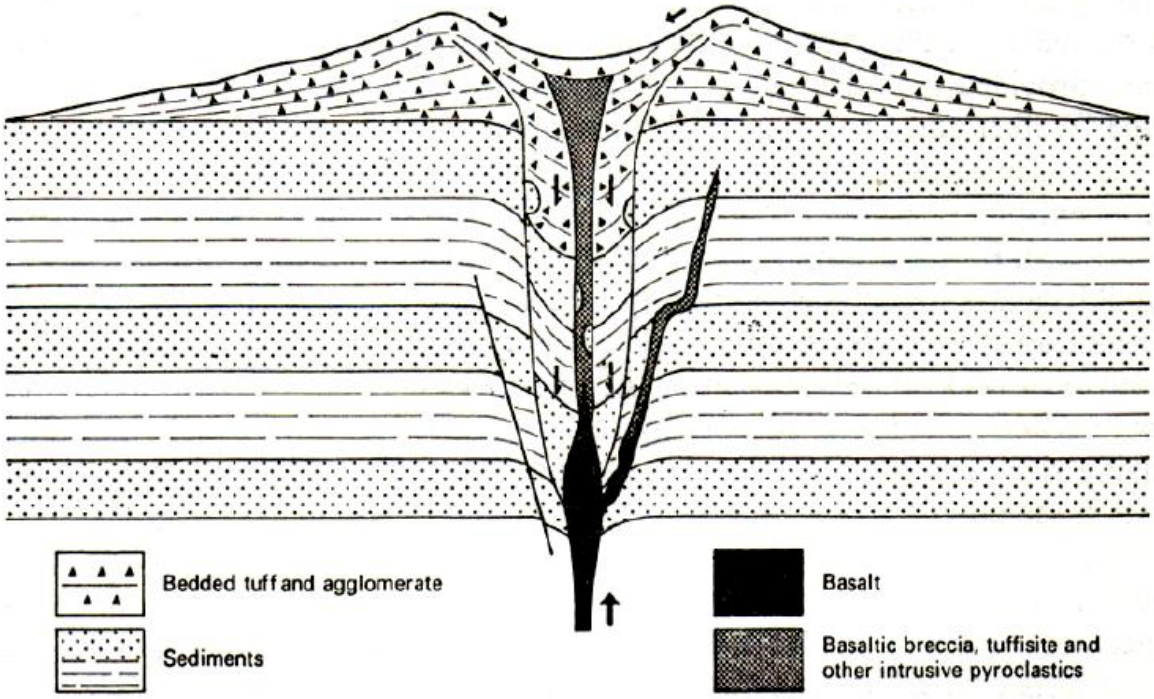


Figure 4.6: Schematic cross-section of a volcanic vent.

4.4 Pittenweem (West Braes)

Access: From Pittenweem Harbour, walk east along the shore road onto the coastal path and continue c. 500 m to the steps down to the tidal pool, which is very obvious. From the car park at Pittenweem West Braes (Crazy Golf), walk down the steps to the tidal pool. Steeply dipping rocks form the rocky wave-cut platform around the tidal pool. The exposures of interest are directly west of the tidal pool, immediately to the left when you come down the steps. These exposures are completely cut off from the access route at high tide.

There is a public toilet at the West Braes car park.

Geological context: The exposed rocks belong to the Sandy Craig Formation (Strathclyde Group) of Asbian (Viséan, Lower Carboniferous) age.

Aim: Develop skills for defining sedimentary facies in fine-grained terrigenous clastic rocks (sandstones and mudstones). Use the facies concept to interpret sedimentary processes and environments during deposition.

Tasks:

1. The sedimentary rocks here are relatively steeply dipping, so determine younging direction.

2. In pairs or threes, compile a bed-scale (that is, at a scale where details within individual beds can be determined) sedimentary log of the succession to the west of the tidal pool. You will be guided to choose an appropriate interval to log. When choosing your logging scale, estimate the likely length of the sedimentary log, and the level of detail you wish to capture. Information to record includes (but is not limited to): bed-scale variation in grainsize; bed geometry; occurrence of sedimentary structures within the bed, on the bed-base (sole structures) and bed tops; palaeoflow/slope indicators; fossil content; colour variations. Each person should make their own log.

3. As you are compiling the log, document the number of distinct lithofacies you recognise, and interpret the specific sedimentary processes associated with each lithofacies. These can be interpreted from sedimentary structures, bed geometry, and fossil content. Individual lithofacies can be given a unique identifier code (e.g. Lithofacies 1, 2, 3; SstRip = Ripple cross-laminated sandstone, SstTX = Tabular cross-bedded sandstone). Be sure to provide a description in your notebook of each individual facies.

4. Look at the gross patterns in which lithofacies occur and stack. Group your lithofacies into associations that represent a suite of sedimentary processes that would occur in a particular sedimentary environment (i.e., facies associations). Fossil data may be able to help you here. You can add your facies association annotations to your logs too. How many different environments do you recognise? Describe the individual facies associations in your notebook.

Summary/review:

1. Compare your sedimentary log with a variety of existing facies models. This is a first-pass way to help interpret the environments of deposition of the succession you are studying. What are the similarities and differences between your succession and existing facies models? No two environments are exactly the same, so you may wish to augment your model to interpret the very specific types of processes and environments you believe led to the deposition of these sediments.

2. Undertake a quantitative reconstruction of the palaeohydraulics of the succession you have documented.

3. Sketch a facies model that incorporates Waltherian principles, juxtaposing your interpreted sedimentary environments and incorporating information such as palaeoflow/slope data.

4. What is the closest modern analogue depositional environment to the one you interpreted?

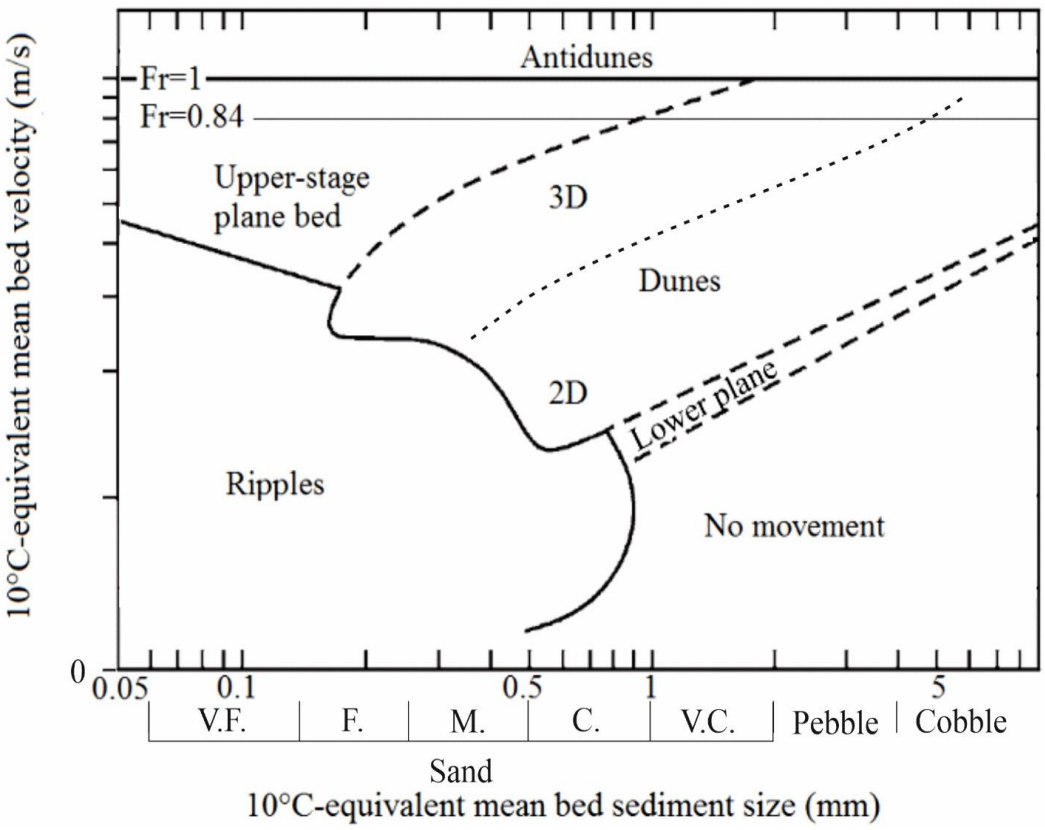


Figure 4.7: The bedform phase diagram (modified from Southard and Boguchwal, 1990).

The Froude number for any flow varies as a function of depth and flow velocity, according to:

$$Fr = \frac{V}{\sqrt{(gD)}}$$

where V = flow velocity (m/s)
g = gravity (9.8 m/s^2)
D = flow depth (m)

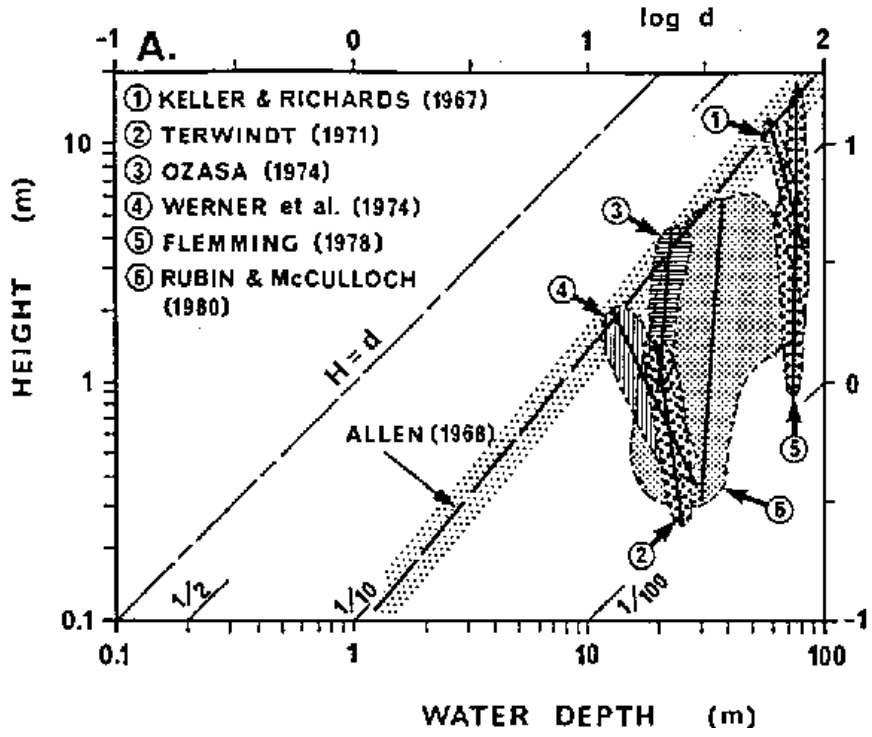


Figure 4.8: The relationship between water depth and dune height (summarised in Flemming, 2000).

4.5 Pittenweem (Pathhead)

Access: From the car park at Pittenweem West Braes (Crazy Golf), walk c. 700 m west along the coastal footpath. From the Pathhead Nurseries access road, take the path to the rocky wave-cut platform and low cliffs. Exposures of interest are directly to the west when accessing the wave-cut platform and low cliffs.

There is a public toilet at the West Braes car park. Toilet facilities may be available at Pathhead Nurseries.

Geological context: The exposed rocks belong to the Pathhead Formation (Strathclyde Group) of Asbian to Brigantian (Viséan, Lower Carboniferous) age. The rocks are stratigraphically higher (younger) than those at the tidal pool.

Aim: Refine skills for defining sedimentary facies in fine-grained terrigenous clastic rocks (sandstones and mudstones). Use the facies concept to interpret sedimentary processes and environments during deposition.

Tasks:

1. The sedimentary rocks here are relatively steeply dipping, so determine younging direction.

2. In pairs or threes, compile a bed-scale (that is, at a scale where details within individual beds can be determined) sedimentary log of the succession to the west of the Pathhead Nurseries access point. You will be guided to choose an appropriate interval to log. When choosing your logging scale, estimate the likely length of the sedimentary log, and the level of detail you wish to capture. Information to record includes (but is not limited to): bed-scale variation in grainsize; bed geometry; occurrence of sedimentary structures within the bed, on the bed-base (sole structures) and bed tops; palaeoflow/slope indicators; fossil content; colour variations. Each person should make their own log.

3. As you are compiling the log, document the number of distinct lithofacies you recognise, and interpret the specific sedimentary processes associated with each lithofacies. These can be interpreted from sedimentary structures, bed geometry, and fossil content. Individual lithofacies can be given a unique identifier code (e.g. Lithofacies 1, 2, 3; SstRip = Ripple cross-laminated sandstone, SstTX = Tabular cross-bedded sandstone). Be sure to provide a description in your notebook of each individual facies.

4. Look at the gross patterns in which lithofacies occur and stack. Group your lithofacies into associations that represent a suite of sedimentary processes that would occur in a particular sedimentary environment (facies associations). Fossil data may be able to help you here. You can add your facies association annotations to your logs too. How many different environments do you recognise? Describe the individual facies associations in your notebook.

Summary/review:

1. Compare your sedimentary log with a variety of existing facies models. This is a first-pass way to help interpret the environments of deposition of the succession you are studying. What are similarities and differences between your succession and existing facies models? No two environment are exactly the same, so you may wish to augment your model to interpret the very specific types of processes and environments you believe led to the deposition of these sediments.

2. What kind of water depths were these sediments deposited in?

3. Sketch a facies model that incorporates Waltherian principles, juxtaposing your interpreted sedimentary environments and incorporating information such as palaeoflow/slope data.

4. What is the closest modern analogue depositional environment to the one you interpreted?

5. What are the differences in interpreted environment between the successions at Pathhead and West Braes? How could you account for these differences?

4.6 St Monans

Access: Park or drop-off at the parking lot at the west end of Braehead, behind St Monans Church. Turn right from the car park onto Burnside towards the coast. Take the Fife Coastal Path westwards past St Monans Church. The exposures of interest are ~250-300 m along the coast and can be accessed either from the wave-cut platform or (at high tide) from the coastal path.

The public toilet in St Monans is a 15-minute walk from the field location. Take the coast path east onto Braehead; continue along Braehead, then cross Station Road onto Hope Place. Follow the road to the right; the public toilets are on the left past the tennis courts and next to the car park.

Geological context: The exposed sedimentary rocks belong to the Pathhead Formation (Strathclyde Group) of Asbian to Brigantian (Viséan, Lower Carboniferous) age. The igneous rocks are of Late Carboniferous to Early Permian age (272-311 Ma).

Aim: Refine skills for describing and measuring igneous rocks on outcrop scale. Interpret mechanisms of igneous intrusion and/or extrusion. Interpret temporal relationship between igneous, sedimentary and structural features.

Task: You have been employed as an exploration geologist by GEMS.co, a dealership of precious gems run from the Manchester area. GEMS.co has learnt that abundant mantle xenoliths have been found in the volcanic necks and plugs of Fife, and believes that this geological occurrence could present an enticing prospect to recover economic quantities of diamonds and other gem-quality igneous minerals.

You are tasked with conducting a geological survey of short stretch of the St Monans coastline. You will present your observations in the form of a compass-and-pace geological map and a survey notebook, and you will use your observational evidence to make recommendations concerning suitable locations to begin exploration and extraction activities. The task is divided into two stages:

Stage 1: Collect primary field data
Stage 2: Interpret field data, answering the three summary questions at the end of this section.

The summary questions ask you to interpret the primary data you collected and recorded in the field, and you will need to draw on your field data (map and notebook) as evidence to justify your interpretations. This means that you will need to prioritize your field activities towards collecting a sufficient quantity and quality of data to be able to address these questions.

In the field:

1. Construct a compass-and-pace geological map of the lithologies and structures exposed along the St Monans coastal section between the Dovecot (round tower) to the west and the Long Shank (prominent rock face extending into the sea) to the east. You will be guided to focus on a particular area. All of the lithologies will be familiar to you from earlier excursions in the field course.

2. Determine the spatial and temporal relationship between the igneous rocks and the surrounding sedimentary rocks. To do this, you will need to produce detailed sketches describing the nature of the contacts between the igneous and sedimentary rocks at a finer scale than can be represented on your map.

3. Produce descriptions and labelled sketches of the igneous rocks, focusing on the rocks you consider most likely to be fruitful for exploration.

Interpretation and summary:

Write a short paragraph to answer each of the following questions:

1. Did the folding of the sediments pre- or post-date the igneous activity? What is the evidence for this? You should refer to numbered pages/sketches in your field notebook, or numbered locations on your map, as evidence to support your answer.

2. Were the igneous rocks intruded into wet/unconsolidated sediment, or into dry and lithified sediment? What evidence is there to support your interpretation? You should refer to numbered pages/sketches in your field notebook, or numbered locations on your map, as evidence to support your answer.

3. Where in the mapped area would you recommend that GEMS.co begins its exploration activities, and why? Justify your recommendation using evidence from your map and notebook.

You may give one or more ranked preferences, if you think there is more than one area that might be suitable for explanation. You may recommend that GEMS.co does not pursue exploration in this area. There is not necessarily a "right" answer to this question – the important aspect is that you are able to reach a decision on you recommendation, and <u>justify</u> your recommendation using the evidence you collected in the field.

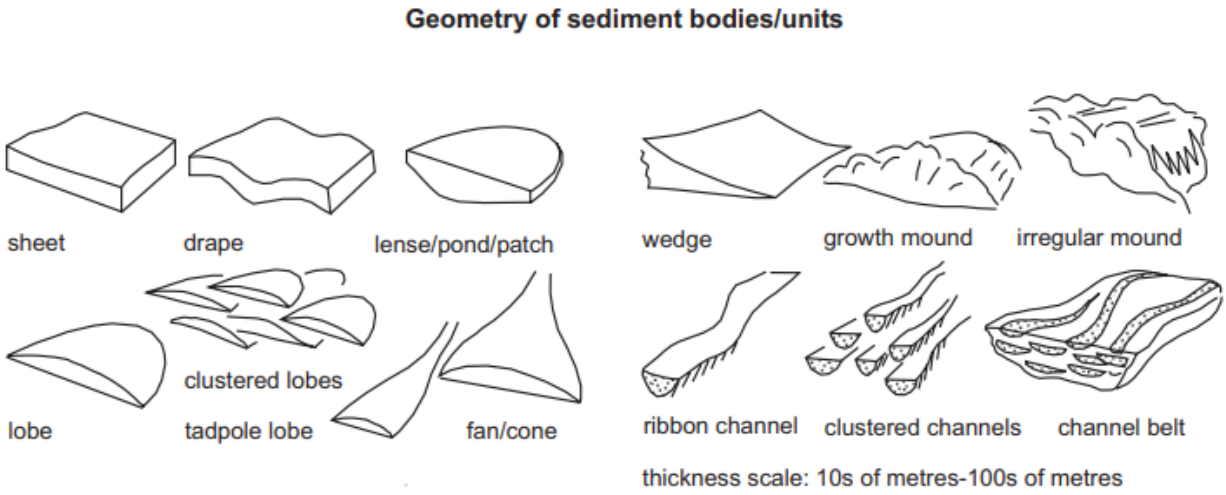
You will probably wish to spend a little time tidying up, inking and colouring in your geological map, but you do not need to add information to your field notebook unless you wish to add short interpretive remarks or thoughts to the work you did in the field.

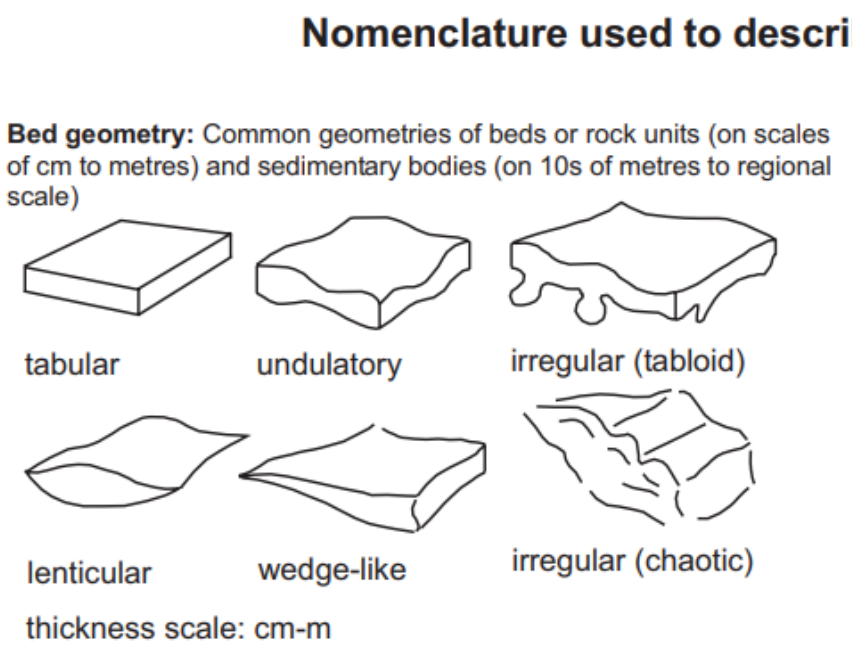
The questions should take no more than two sides of A4 paper to answer in full. Do not spend time copying out observations or sketches from your notebook – simply write your answer to the question, and refer to your map or notebook as evidence when needed.

Your answers to the three summary questions should be handed in, along with your map and notebook, at the end of the field course.

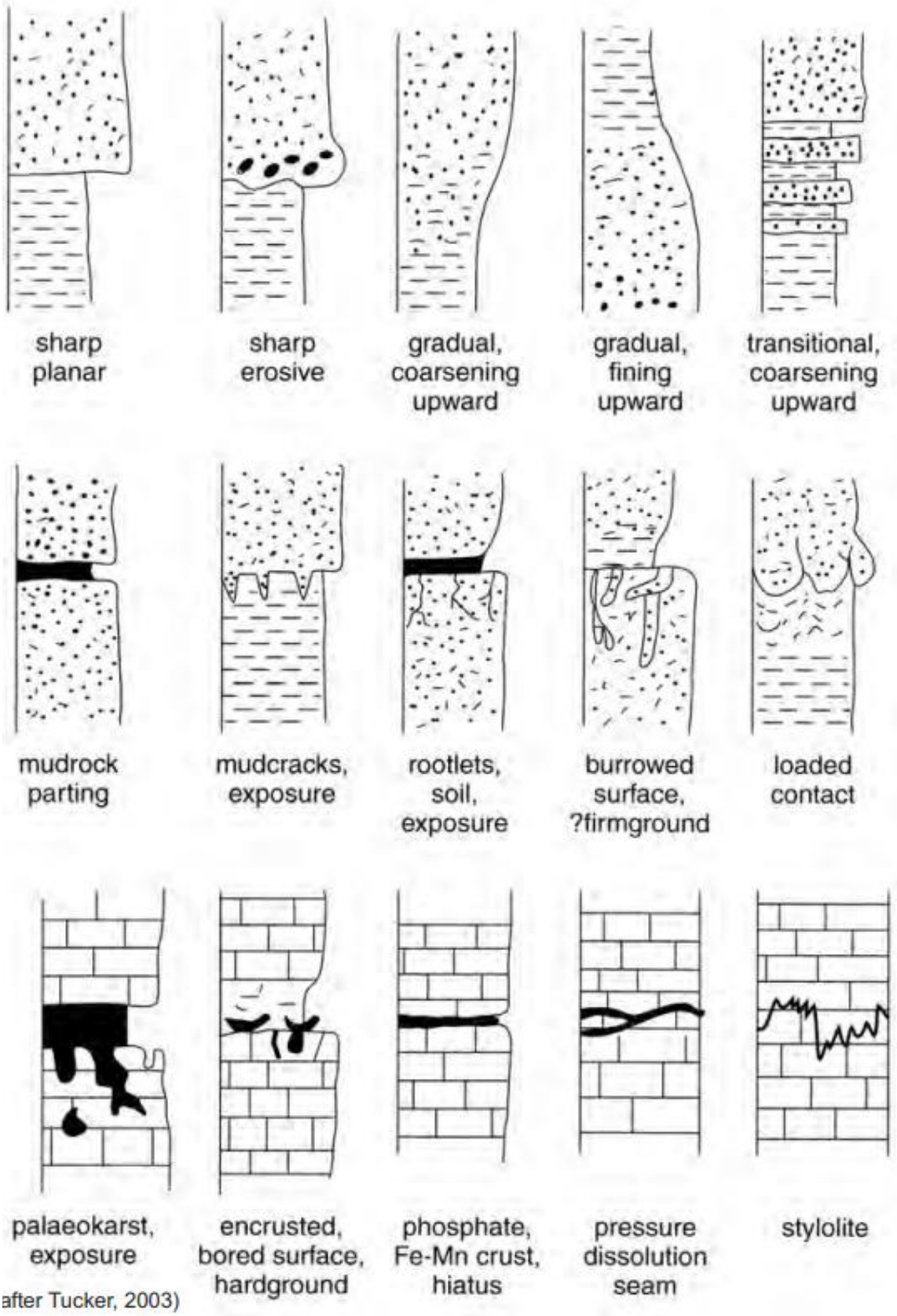
Part 5 - Useful resources

5.1 Describing sedimentary rocks in the field





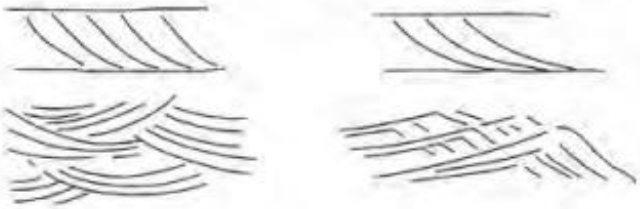
Terminology for bedding trends and contacts



after Tucker, 2003)

Useful structures for determining the way up of strata

cross-stratification (all scales: cross bedding, cross lamination, etc)

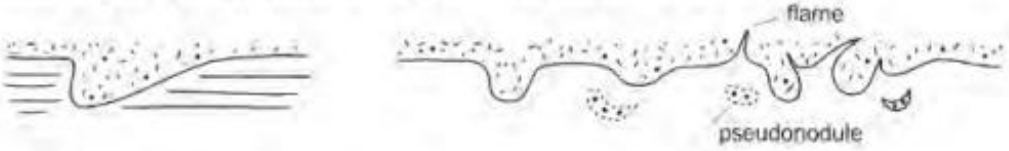


truncated tops, curved bases

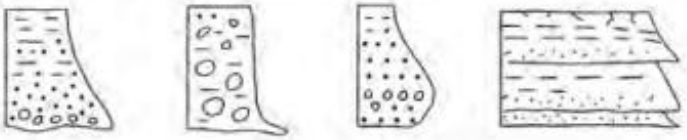
erosional structures (all scales: channel scours, micro scours, etc)



sole marks (grooves, tools, flutes, loads and associated features)



grading (graded beds/laminae) and structural sequences (Bouma, Stow, etc)



normal grading common but BEWARE because reverse / symmetrical grading also occurs!! Look for erosional scour at base.

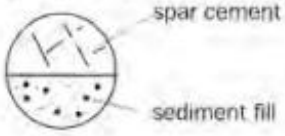
surface marks (ripples, desiccation cracks, rain drops, lineation, footprints, etc)



water-escape structures (dishes, burst-throughs, sand volcanoes)



geopetals



grazing trails (bed surface)



fossils in growth position eg.stromatolite



(after Stow, 2005)

Classification of siliciclastic rocks

Most usefully described in terms of grain size and grain composition.
Textural properties (such as grain morphology and sorting) are also described.

Length (mm)	φ		Grain size descriptor	Sediment/ rock name
			block	mega-conglomerate
4096	-12	vc	boulder	gravel conglomerate (rounded clasts) breccia (angular clasts)
2048	-11	c		
1024	-10	m		
512	-9	f		
256	-8	c	cobble	
128	-7	f		
64	-6	vc	pebble	
32	-5	c		
16	-4	m		
8	-3	f		
4	-2		granule	
2	-1	vc	sand	sand sandstone
1	0	c		
0.50	1	m		
0.25	2	f		
0.125	3	vf		
0.063	4	c	silt	silt siltstone
0.031	5	m		
0.015	6	f		
0.008	7	vf		
0.004	8		clay	clay claystone

• Use the Udden-Wentworth scale

Grain size typically reported in terms of three main divisions:

- Gravel/conglomerate (>2.00 mm)
- Sand/sandstone (<2.00 mm to >0.063 mm)
- Mud/mudstone (<0.063 mm)

Suffix the term -stone on the grain-size descriptor to imply the sediment has been lithified (except gravel which becomes conglomerate or breccia when lithified depending on the angularity of the clasts).

Sedimentary rock composed of:

- Framework grains (residual components of weathering i.e. quartz, feldspar and lithic fragments)
- Non-framework component
  - matrix
  - diagenetic pore filling components (e.g. cement)

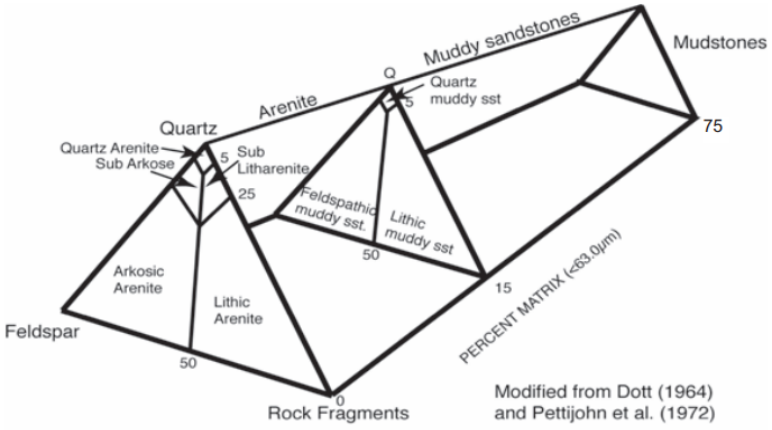
Definition of matrix:

- Conglomerates: <2 mm
- Sandstones: frequently very fine material <30 μm, (Dott, 1964) but <63 μm is also used as this is mud/sand boundary
- Be wary about the origin of clay: use textural analysis to identify an authigenic clay component masquerading as matrix

Composition of sandstones

Grain composition is described in terms of varying proportions of quartz, feldspar, and rock fragments (lithic grains):

- "Arenite": clean sand(stone) (>50% sand sized grains) containing <15% matrix
- "Wacke" or "Muddy sandstone": dirty sand(stone) (>50% sand sized grains) containing >15% and < 75% matrix. Sometimes called a greywacke.
- "Quartz arenite": arenite containing >95% quartz grains
- "Arkosic arenite": arenite containing >25% feldspar and in which the amount of feldspar is greater than that of rock fragments
- "Litharenite": arenite containing >25% rock fragments and in which the amount of rock fragments is greater than that of feldspar
- Put together in 3D, a typical sandstone classification is:



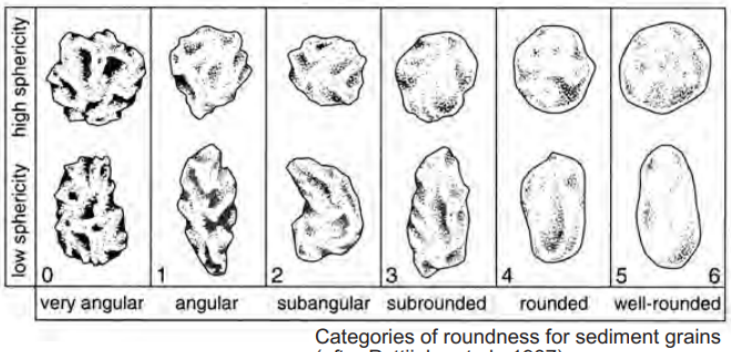
Modified from Dott (1964) and Pettijohn et al. (1972)

Grain morphology: shape, sphericity and roundness

Definitions:
Shape: defined by the ratio of long, intermediate and short axes of the grain
Sphericity: a measure of how closely the grain shape resembles a sphere
Roundness: a description of the smoothness of the corners of the grain

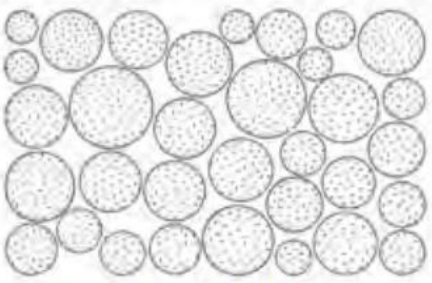
Clast shape is generally a function of:
- the composition of the source material
- any planes of weakness in the rock (e.g. bedding/lamination, cleavage or jointing)

The clast morphology is not generally a reliable indicator of depositional environment, although a population of very well rounded grains does suggest either:
- many cycles of transport or
- intense abrasion in a special environment (e.g. a desert or beach)

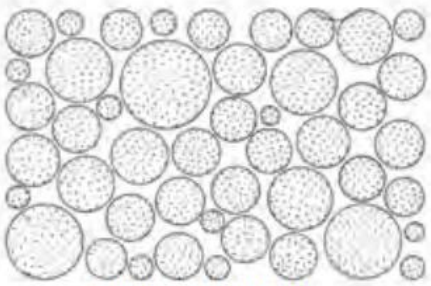


Categories of roundness for sediment grains (after Pettijohn et al., 1987)

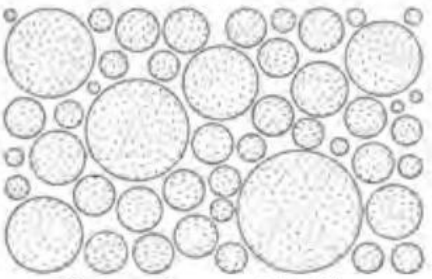
Quantification of textural properties: sorting



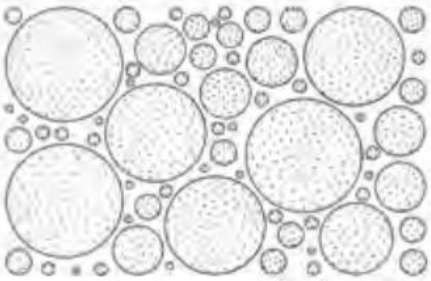
Very well sorted (0.35)



Well sorted (0.50)



Moderately sorted (1.00)

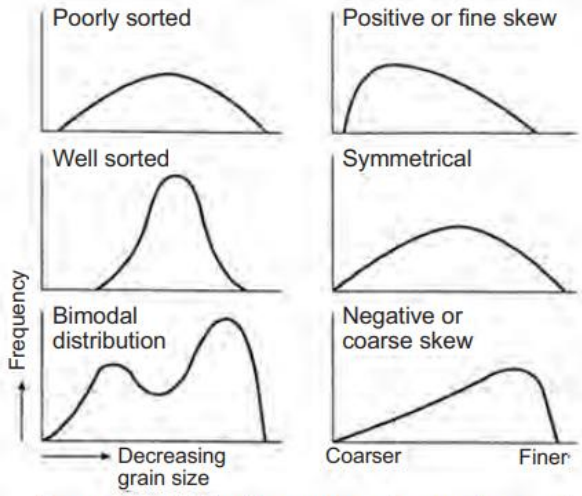


Poorly sorted (2.00)

Diameter Ratio* (mm)	PHI Standard Deviation	Verbal Scale	
1.0	0.00		
		very well sorted	Mature
1.6	0.35		
		well sorted	
2.0	0.50		
		moderately sorted	Submature
4.0	1.00		
		poorly sorted	
16.0	2.00		
		very poorly sorted	

* ratio between the largest and smallest grain

(after Folk, 1965)



Frequency distribution curves showing types of sorting and skewness

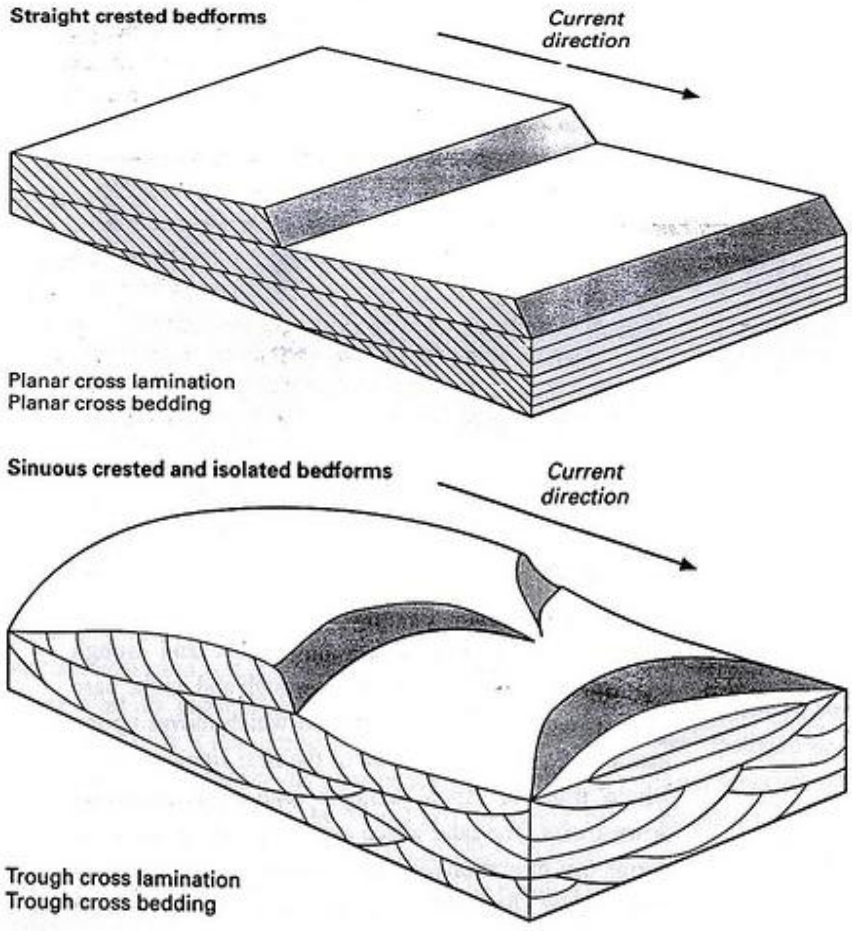
Controls on sorting:
- source rocks
- grain size
- depositional mechanism
- distance transported

Sorting and skewness are of dubious value in *determining depositional environments*

(after Folk, 1965)

Relationship between bedforms and flow properties

	Ripples	2D dunes	3D dunes	Lower plane bed	Upper plane bed	Antidunes
Length (spacing)	0.1–0.2 m	a few 10s of cm to 100s of m	a few 10s of cm to 10s of m (or more?)	—	—	10s of cm–m
Height	a few cm	cm to a few 10s of m	10s of cm to a few m (or more?)	—	—	cm–10s of cm
Ripple index (length/height)	relatively low	relatively high	relatively low	—	—	relatively high
Plan geometry	strongly irregular/ short-crested	straight/ sinuous and long-crested	strongly irregular/ short-crested	—	—	long-crested and short-crested
Characteristic flow velocity	low	low/moderate	moderate/high	low	high	high
Characteristic flow depth	> a few cm	> a few dm	> a few cm	all	all	shallow lows
Characteristic sediment size	0.03–0.6 mm	>0.3 mm	>0.2 mm	>0.6 mm	all	all



Cross-bedding patterns generated by straight-crested (“2D”) and lunate-to-sinuous (“3D”) dunes, and how to determine flow orientation from them.

5.2 Describing igneous rocks in the field

Plutonic rocks

Describe the **contacts**:

- Nature of the contact: concordant/discordant, sharp/gradational
- Shape and orientation of the contact: planar, non-planar, strike/dip
- Any fabrics, foliations, changes in grain size, or xenoliths in the igneous body near the contact
- Any disturbance of the country rock, e.g. baking, fracturing

Look for observations that help constrain the depth and/or mode of emplacement, e.g. stoping, diapirism, hydraulic fracturing.

Describe any **internal compositional or fabric variations**:

- Nature of variation, e.g. layering, mixing
- Spatial and compositional scale of variation. Use graphic logs if appropriate

Look for observations that help constrain the origin of the variations, e.g. cumulate crystal settling

Describe any **internal boundaries or internal contacts** that indicate different phases of intrusion:

- Describe the contacts (as above)
- Look for evidence of relative ages, e.g. cross-cutting relationships, chilled margins

Describe any **jointing**.

Hypabyssal rocks – rocks that solidified in the upper part of the crust, usually as sills or dykes

Describe the **contacts**:

- Nature of the contact: concordant/discordant, sharp/gradational
- Shape and orientation of the contact: planar, non-planar, strike/dip
- Thickness of the intrusion
- Offsets of surfaces in the country rock across the intrusion

Look for observations that help constrain the depth and/or mode of emplacement.

Describe any **internal compositional or fabric variations**:

- Chilled margins
- Presence of within-flow differentiation such as phenocryst layering
- Flow banding; preferred orientation of phenocrysts
- Vesicles or amygdales

Describe any **jointing** (e.g., columnar jointing) and measure its orientation and spacing.

Volcanic rocks

Volcaniclastic sequences can be described using the same methods as sedimentary sequences. The easiest way to do this is to record up-stratigraphy variations on a measured section (either on logging paper or in your notebook), and to record lateral variations by constructing more than one measured section.

Describe the **contacts:**

- Orientation of the contact with respect to the underlying country rock
- Nature of the contact: sharp/gradational, transitional, loaded, paleosol horizons, etc.

Describe the **geometry of the units:**

- Shape of the unit, e.g. tabular, lenticular, wedge-like
- Thickness of the unit (and note any lateral thickness variations)

Describe any **variations within units:**

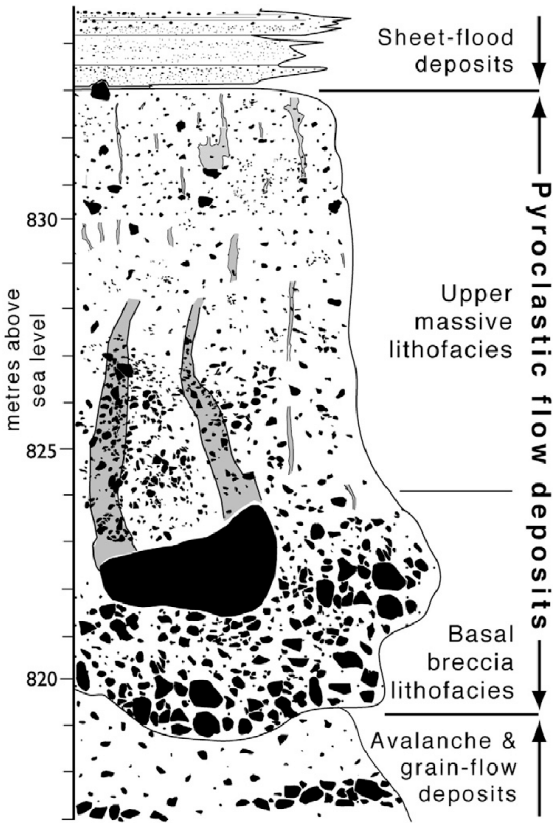
- Changes in grain size: normal grading, reverse grading
- Changes in fabric: laminations, cross-laminations, flow banding
- Changes in composition: abundance and properties of lithics, pumice, fiamme, crystals

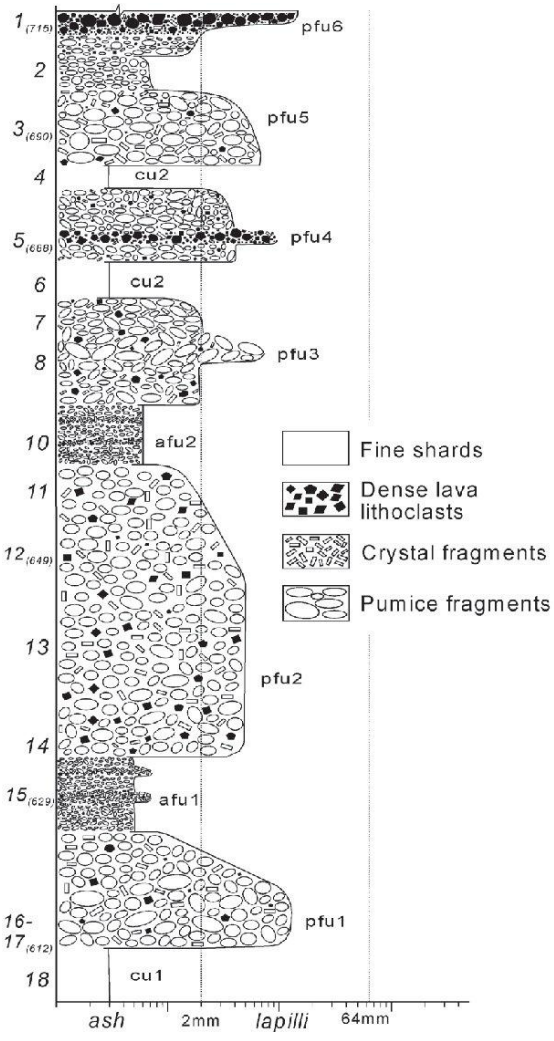
Describe any **jointing, fracturing, veining, mineralization**.

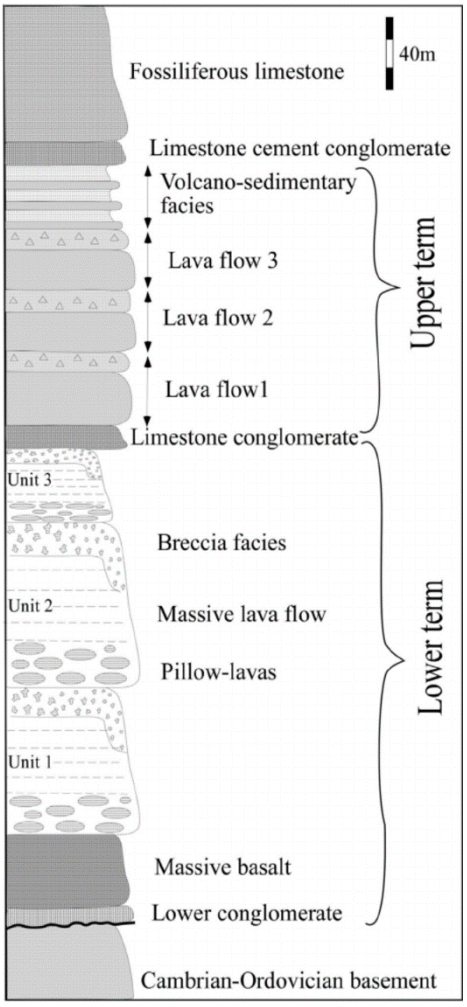
Grain size	Volcanic clast type	Sedimentary equivalent
>25.6 cm	coarse blocks and bombs	boulders
>6.4 cm	fine blocks and bombs	cobbles
>2 mm	lapilli	gravel
>1/16 mm	coarse ash	sand
< 1/16 mm	fine ash	silt

Lava flows can be included on measured sections, and many of the same features should be described (e.g., contacts, flow thickness, internal variations within the flow). Remember that lava flows are NOT sedimentary rocks: they are *emplaced*, not deposited.

The following page shows three examples of measured sections through volcanic sequences. Each has a slightly different style, but each diagram conveys essential information about the nature of the volcaniclastic deposits and/or lava flows in the sequence.


Sheet-flood deposits
Pyroclastic flow deposits
Upper massive lithofacies
Basal breccia lithofacies
Avalanche & grain-flow deposits
metres above sea level
830
825
820

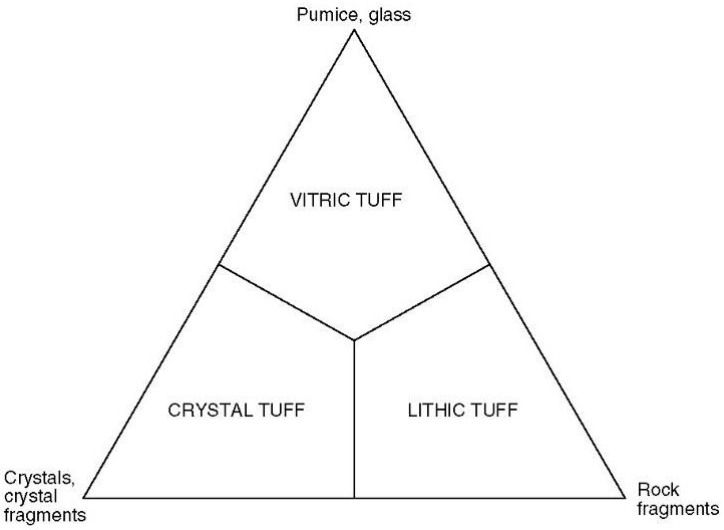

1 (715)
2
3 (650)
4
5 (668)
6
7
8
10
11
12 (649)
13
14
15 (629)
16-17 (612)
18
pfu6
pfu5
cu2
pfu4
cu2
pfu3
afu2
pfu2
afu1
pfu1
cu1
Fine shards
Dense lava lithoclasts
Crystal fragments
Pumice fragments
ash
2mm
lapilli
64mm


40m
Fossiliferous limestone
Limestone cement conglomerate
Volcano-sedimentary facies
Lava flow 3
Lava flow 2
Lava flow1
Limestone conglomerate
Upper term
Unit 3
Breccia facies
Unit 2
Massive lava flow
Pillow-lavas
Unit 1
Massive basalt
Lower conglomerate
Lower term
Cambrian-Ordovician basement

Pyroclastic rocks have fragmentary textures resulting from explosive igneous activity. The main components of pyroclastic rocks are:

a) Volcanic glass and pumice
b) Crystals and crystal fragments
c) Lithic (rock) fragments, i.e. xenoliths and xenocrysts.

The proportions of glass, crystals and lithics can be used in classification.



There are three ways to classify pyroclastic rocks:

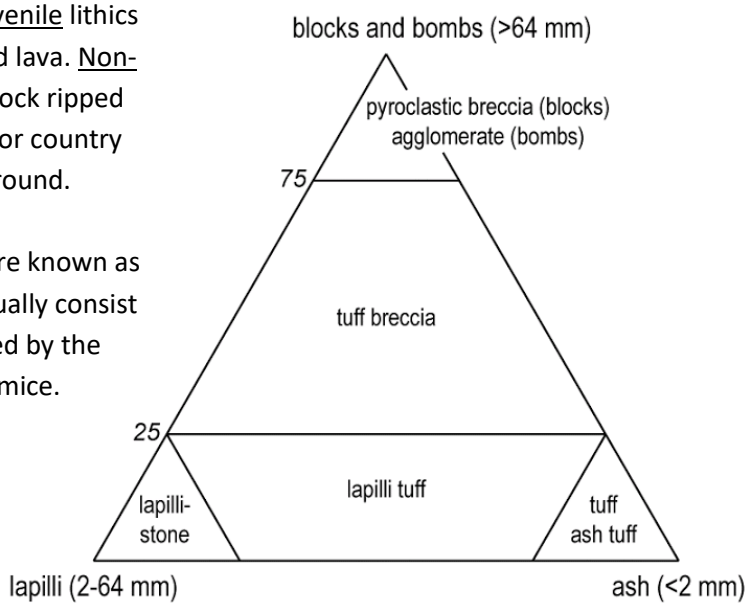
1. Petrological: i.e. basaltic tuff, rhyolitic tuff

2. Lithological: this is based on the proportions of the various components, and the grain size.

Lithified or consolidated pyroclastic rocks are generally known as agglomerates when composed of blocks and bombs, or tuffs when composed largely of lapilli and ash-sized grains.

Lithic fragments are non-vesicular. Juvenile lithics derive from the disruption of solidified lava. Non-juvenile lithics comprise old volcanic rock ripped up from the vent wall, xenoliths, and/or country rock picked up from the vent or the ground.

Vesicular and lapilli-sized fragments are known as **pumice**. Fine-grained **ash** particles usually consist of free crystals and glass shards formed by the fragmentation of frothy magma or pumice.



3. Genetic classification of tephra deposits

Pyroclastic deposits may be classified according to their formation into: **pyroclastic fall** deposits or **pyroclastic density current** deposits (also called pyroclastic flow deposits). This is the most interesting approach, but if the rock is wrongly classified it can lead to problems interpreting the volcanic processes that formed the rock.

Pyroclastic fall deposits contain pyroclasts that are coarsest near the vent (bombs and coarse lapilli) and become increasingly finer-grained away from the vent (fine lapilli and ash). The thickness and dispersal of the pyroclasts depends on the type and magnitude of the eruption that formed the deposit. Two end-member genetic types of fall deposits are recognised:

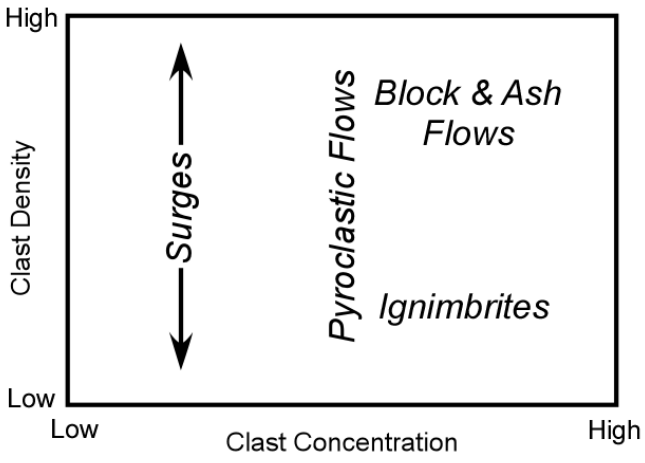
Scoria fall deposits derive from Strombolian eruptions of scoria cones. Composed of basaltic to andesitic vesiculated pyroclasts (scoria). Deposits do not usually spread far from the vent.

Pumice fall deposits derive from Plinian eruptions of stratovolcanoes. Composed of highly vesiculated dacitic to rhyolitic pyroclasts (pumice). Deposits may be distributed for hundreds of square kilometres away from the vent.

Pyroclastic density current deposits (also called pyroclastic flow deposits) are classified according to the density of the individual pyroclasts, and the concentration of the current that carried them.

Ignimbrites are the classic "pyroclastic flow deposit". They have distinctive ignimbritic textures that can be recognised in hand specimen and thin section. Many ignimbrites are strongly compacted and welded, so that the rock takes on a preferred orientation of flattened pumice lumps (fiammé) and/or stretched-out shards, resulting in a eutaxitic texture. Most ignimbrites are rhyolitic or dacitic in composition.

Ignimbrite deposits tend to be massive and poorly sorted. Pyroclastic surge deposits often exhibit cross-bedding.



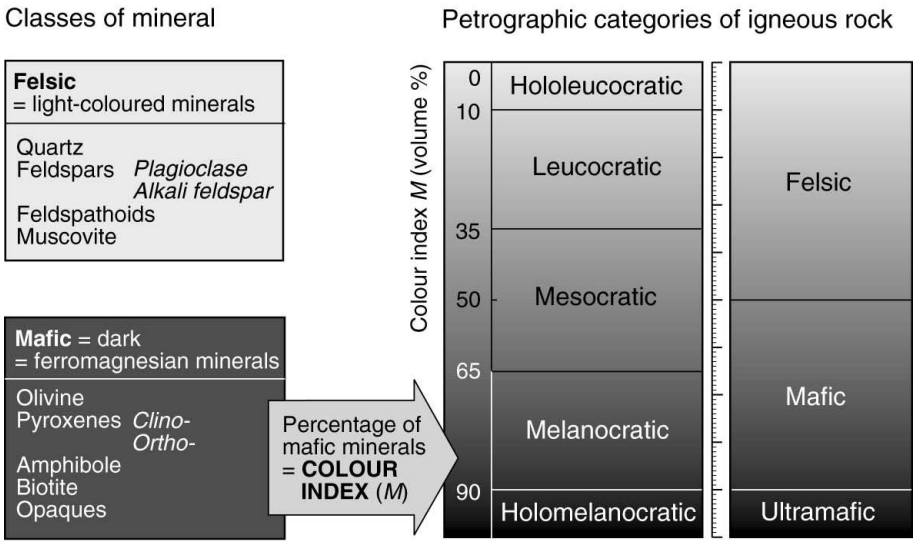
Igneous rocks in hand specimen

The aim of a hand-specimen description is to provide enough information to name or classify the igneous rock.

Remember that a weathered surface may be just as informative as the freshly broken surface. Sometimes weathered surfaces show up features that are not easily visible on a fresh surface.

1. Colour

Estimate the colour index. This may be easier with a thin section to hand. The colour index is given by the total modal content (i.e. volume %) of ferromagnesian minerals (olivine, pyroxenes, amphibole, biotite).



Gill (2010) Fig. 1.3b

2. Crystallinity and grain size

Igneous rocks range from being completely crystalline (holocrystalline) to completely glassy (holohyaline, glassy or vitreous), or may be partially crystalline and partly glassy (hypohyaline or hypocrystalline). Grain size terminology is summarized below:

Average mineral diameter	Textural terms
>30 mm	pegmatitic
5-30 mm	very coarse-grained
3-5 mm	coarse-grained (groundmass crystals can be identified with the naked eye)
1-3 mm	medium-grained (individual grains are visible with the naked eye, but too small to identify)
0.05-1 mm	fine-grained (individual crystals are too small to see with the naked eye)
<0.05 mm	very fine-grained

3. Presence or absence of phenocrysts

Use the term porphyritic if phenocrysts are present, or even-grained if the rock is crystalline but has no phenocrysts. For lavas or glassy rocks, use aphyric for rocks with no phenocrysts, or "XX-phyric" for a rock with phenocrysts of mineral XX.

> The above three definitions can usually be written down in a single sentence. For example:
>
> "This melanocratic rock is holocrystalline, medium-grained, and is not porphyritic."

4. Structure and texture: describe any banding, lineation, jointing, vesicularity, inclusions, xenoliths, and anything else of interest.

5. Identify all the minerals present as far as possible. Make a note of any special features of the minerals (e.g. obvious zonation or cleavage), or of the whole rock (e.g. state of alteration).

Record the sizes and modal abundances of each mineral. Record the modal proportions of phenocrysts and groundmass.

For each mineral, record the following information (this will be easier in coarser-grained rocks):

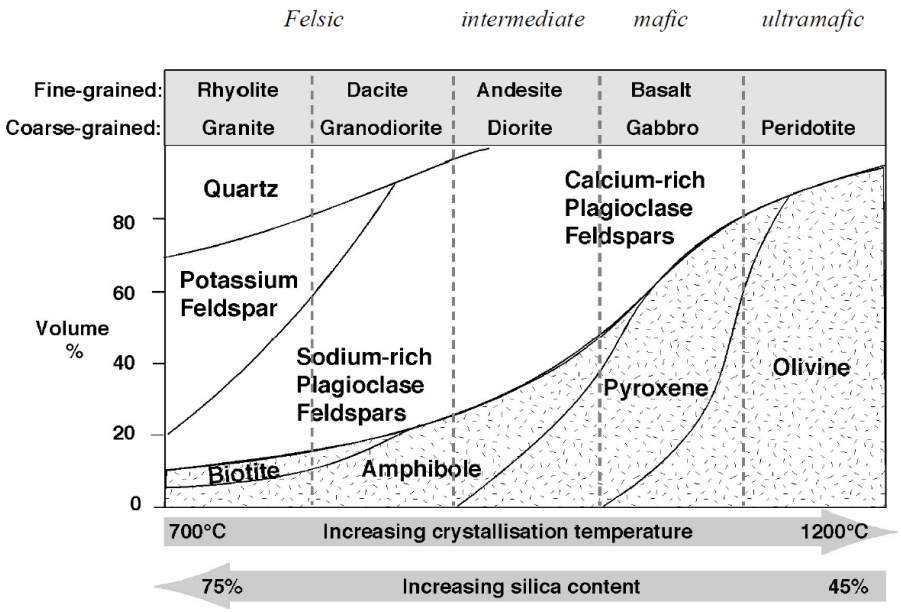
- Basis for identification: colour, hardness, habit, cleavage, twinning
- Crystal size and shape e.g. euhedral, subhedral, anhedral
- Textural relationships with other minerals, e.g, groundmass phase, phenocryst phase, poikilitic.

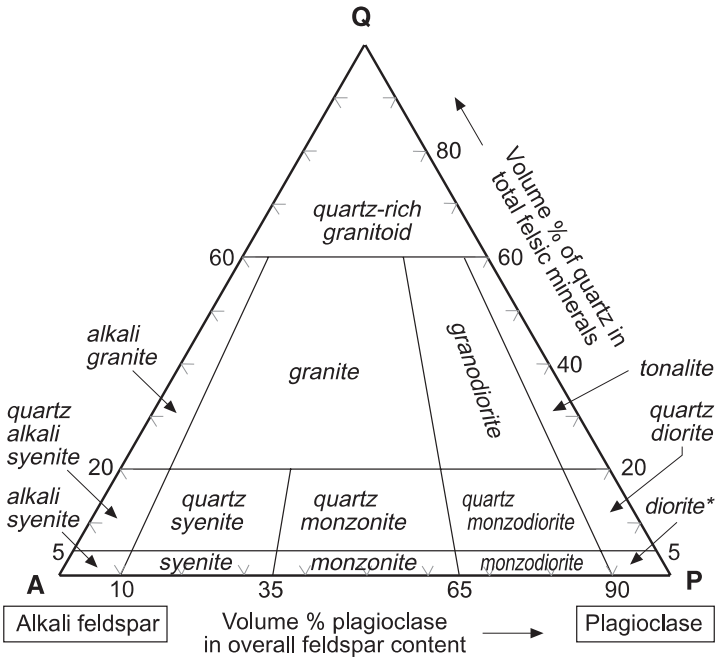
Use mineral associations to predict which other minerals you may have missed.

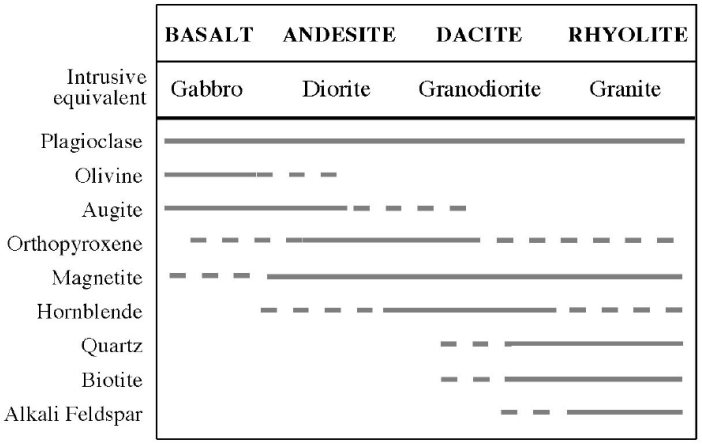
6. Classify the rock: Give the rock a field name. In many cases it will only be possible to use a simple classification scheme.

Fine-grained or volcanic rock	**Medium or coarse-grained equivalent**	**Common phenocryst minerals** (this list is neither exhaustive nor prescriptive)
	Felsic rocks	
rhyolite	granite	quartz, alkali feldspar
dacite	granodiorite	quartz, Na-rich plagioclase, alkali feldspar
trachyte	syenite	alkali feldspar
phonolite	nepheline syenite	alkali feldspar, nepheline or leucite, aegirine/augite
	Mafic rocks	
andesite	diorite	Na-rich plagioclase, pyroxene or amphibole
basalt	gabbro	Ca-rich plagioclase, pyroxene, olivine

Fine-grained rock types, their coarse-grained equivalents, and common phenocryst minerals







Mineral associations in igneous rocks associated with subduction zone magmatism

Part 6 - References and bibliography

Online resources

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Especially:

- Chapter 5: Bluck BJ (2002) The Midland Valley Terrane, pp 149-166
- Chapter 8: Trewin NT, Thirlwall MF (2002) The Old Red Sandstone, pp 213-250
- Chapter 9: Read WA, Browne MAE, Stephenson D, Upton BJG (2002) Carboniferous, pp 251-300

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